2.16. Да се докаже, че $\forall$ крайна група G от ред 8 е изоморфна точно на една от групите D_4 или Q_8.

$|G|=8 \qquad |G| = |H| \cdot |G:H| \Rightarrow \forall a \in G : |a| \mid |G|$

- ако $\exists g \in G : |g| = 8 \Rightarrow \langle g \rangle = G$ – цикл. $\Rightarrow$ абел. ↯

- G е от четен ред $\Rightarrow \exists g \in G : |g| = 2$

- ако $\forall$ (четен ред) на G е от ред 2 $\Rightarrow G$ е абел ↯

$\Rightarrow \exists a \in G : |a| = 4$

$b \in G \setminus \langle a \rangle$, $|G : \langle a \rangle| = \frac{8}{4} = 2 \Rightarrow G = \langle a \rangle \cup b\langle a \rangle \Rightarrow G = \langle a, b \rangle$

ако $ab \in \langle a \rangle \Leftrightarrow ab = a^k$, $k = 0,1,2,3 \Leftrightarrow b = a^{k-1}$ ↯

$\Rightarrow ab \in b\langle a \rangle \Leftrightarrow ab = ba^k \Leftrightarrow b^{-1}ab = a^k$, $k = 0, \dots, 3$

$ab = ba \Rightarrow G$ е абел. ↯

Знаем, че $|b^{-1}ab| = |a| = 4 \Rightarrow |b^{-1}ab| = 4 \Rightarrow b^{-1}ab = a$ или

$\boxed{b^{-1}ab = a^3 = a^{-1}}$ ✓

От кой ред е b?

1сл. $|b| = 2 \Rightarrow G = \langle a, b \mid |a| = 4, |b| = 2, b^{-1}ab = a^{-1} \rangle \cong D_4$

2сл. $|b| = 4$

$b^2 \in b\langle a \rangle \Leftrightarrow b^2 = ba^k \Leftrightarrow b = a^k$ ↯

$\Rightarrow b^2 \in \langle a \rangle \Leftrightarrow b^2 = a^k$ и $|b^2| = 2 \Rightarrow b^2 = a^2$

$ab = c \Rightarrow c^2 = \underbrace{ab}_{a^3b}\underbrace{ab}_{a^4} = \underline{aa^3}bb = b^2 = a^2 \Rightarrow |c| = 4$

$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$

$i^2 = j^2 = k^2 = -1$ ✓

$ij = k$, $ji = -k$ ✓

$jk = i$, $kj = -i$ ✓

$ki = j$, $ik = -j$ ✓

$b^3ab = a^3 \mid \cdot b$ – отляво $\Rightarrow ab = b a^3$

$b^3a^2b = a^2 \Rightarrow a^2b = ba^2$

$b^3a^3b = a \Rightarrow a^3b = ba$

$c = ab \overset{?}{=} a^2ba$

$a^2\underbrace{ba}_{a^3b} = \underbrace{a^2a^3}_{a}b = ab = c$ ✓

$bc = \underbrace{ba}_{a^3b}b = a^3b^2 = a^2ab \cdot b = a^2 \cdot cb$ ✓

$ca = \underbrace{aba}_{a^3b}a = aa^3b = a^2aab = a^2ac$ ✓

16.04.

Теория на числата

$a, b \in \mathbb{Z}$: $a \mid b$, ако $\exists c \in \mathbb{Z}$: $b = ac$

$a \neq 0$

Св-ва: $a \mid b$, $b \neq 0 \Rightarrow |a| \leq |b|$

Зад. Да се докаже, че за $\forall n \in \mathbb{N}$ $a_n = 3^{3n+3} - 26n - 27$ се дели на 26^2

Инд. по n:

1) $n = 1$: $a_1 = 3^6 - 26 - 27 = 27(27-1) - 26 = 26 \cdot 26 = 26^2$ ✓

($3^6 = 27^2$; $27 - 1 = 26$)

2) $26^2 \mid a_k$, $k > 1$

3) $a_{k+1} = 3^{3(k+1)+3} - 26(k+1) - 27 = 3^{3k+6} - 26k - 27 - 26 =$

$3^3(\underbrace{3^{3k+3} - 26k - 27}_{26^2 \mid a_k}) + \underbrace{(3^3-1)}_{26}26k + \underbrace{(3^3-1)}_{26}27 - 26 = \underbrace{3^3 \cdot a_k}_{26^2 \mid} + \underbrace{26^2k}_{26^2 \mid} + \underbrace{26(27-1)}_{26^2 \mid}$ ✓

Д.Р.

5) $a_n = (n+1)(n+2)\ldots(n+n)$ се дели на 2^n, но не се дели на 2^{n+1}

$(k+1)(k+2)(k+3)\ldots(k+k)$ ✓

$(k+1+1)(k+1+2)(k+1+3)\ldots(k+1+k-1)(k+1+k)(k+1+k+1) = (k+2)(k+3)(k+4)\ldots(k+k)(2k+1)(2k+2) = a_k \cdot \underbrace{2(2k+1)}_{2(k+1)}$

б) $a_n = 2^{2^{n+1}} + 2^{2^n} + 1$ се дели на 21

$2^k \mid a_k \Rightarrow 2^{k+1} \mid a_k \cdot 2 \Rightarrow 2^{k+1} \mid a_{k+1}$
$4 \mid a_k \Rightarrow 2^{k+2} \mid a_k \cdot 4$
$2^{k+1} \mid a_k \Rightarrow 2^{k+2} \mid a_k \cdot 2$ $\Rightarrow$ ✓

Зад: ДСК $\forall x \in \mathbb{Z}$, за които $A = x^3 - 8x^2 + 2x$ се дели на $x^2 + 1 = B$

$$\begin{array}{r} x^3 - 8x^2 + 2x \\ -\,x^3 \qquad + x \\ \hline -8x^2 + x \\ -\,-8x^2 \quad -8 \\ \hline x + 8 = r \end{array} \qquad \begin{array}{|l} x^2+1 \\ \hline q = x - 8 \end{array}$$

$A = Bq + r \Rightarrow B \mid r$

1сл. $r = 0 \Leftrightarrow \boxed{x = -8}$

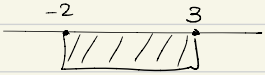
2сл $r \neq 0 \Leftrightarrow B \mid r$

$x^2+1 \mid x+8 \Rightarrow |x^2+1| \leq |x+8|$
$\#0 \qquad\qquad x^2+1$

$\Rightarrow -x-8 \leq x^2+1 \leq x+8$

2.1. $x^2 + x + 9 \geq 0$
$D = -1 - 36 \leq 0$
$\forall x$ е реш.

2.2 $x^2 - x - 7 \leq 0$
$D = 1 + 28 = 29$
$x_{1,2} = \frac{1 \pm \sqrt{29}}{2} \to \approx -2$ / ≈ 3



$\Rightarrow x = -2, -1, 0, 1, 2, 3$

$x^2+1 \mid x+8$
$x = -2 \Rightarrow 5 \mid 6$ ↯
$\boxed{x = 0} \Rightarrow 1 \mid 8$ ✓
$\boxed{x = 2} \Rightarrow 5 \mid 10$ ✓

Т-ма за деление с ост: $\forall a, b \in \mathbb{Z}, b \neq 0$ $\exists!\, q, r \in \mathbb{Z}: a = b \cdot q + r, 0 \leq r < |b|$
(q: частно, r: остатък)

$a = -25, b = 7, q = ?, r = ?$ $\quad -25 = (-4) \cdot 7 + 3$
$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\quad q \quad r$

$\forall a \in \mathbb{N}$, $p \in \mathbb{N}$, $p \geq 1$ $\exists!$ представяне $a = c_k \cdot p^k + c_{k-1} \cdot p^{k-1} + \ldots + c_1 p + c_0$, $c_i \geq 0, i = 0, \ldots, k-1$

p-ично представяне на ест. число $c_k > 0$

Едно число $a \in \mathbb{N}$ се дели на 3 $\Leftrightarrow$ сумата на цифрите му се дели на 3 $\Leftrightarrow$

НОД: $\forall a, b \in \mathbb{Z} : (a,b) \neq (0,0)$

$\exists! d \in \mathbb{N}$ - НОД: $(a,b) = d \geq 1$, ако

1) $d \mid a, d \mid b$

2) $\forall d_1 \in \mathbb{Z} : d_1 \mid a$ и $d_1 \mid b \Rightarrow d_1 \mid d$

$(a,b) = 1$ - a и b са взаимно прости

$(a_1, a_2, \ldots, a_k) = ((a_1, \ldots, a_{k-1}), a_k)$

Алг. на Евклид за нам. на НОД

$a > 0, b > 0$: $a = b \cdot q_1 + r_1$ $\quad 0 \leq r_1 < b$

ако $r_1 = 0 \Rightarrow (a,b) = b$

ако $r_1 \neq 0 \Rightarrow b = r_1 \cdot q_2 + r_2$, $a \leq r_2 < r_1$

ако $r_2 = 0 \Rightarrow (a,b) = (b, r_1) = r_1$

ако $r_2 \neq 0 \Rightarrow$ — $\mathcal{N}$ —

! краен процес

НОД = посл. $\neq 0$ остатък

Теорема на Безу: $\forall a, b \in \mathbb{Z} : \exists u, v \in \mathbb{Z} : ua + vb = (a,b)$

При това $(u,v) = 1$

Зад: $a = 232$, $b = 108$? НОД и u, v

$232 = 2 \cdot 108 + 16$

$108 = 6 \cdot 16 + 12$

$16 = 1 \cdot 12 + 4$
$12 = 3 \cdot 4 + 0$ $\Rightarrow (232, 108) = 4$

$4 = 16 - 1 \cdot 12 = 16 - 1 \cdot (108 - 6 \cdot 16) = 7 \cdot 16 - 1 \cdot 108 = 7(232 - 2 \cdot 108) - 1 \cdot 108 = 7 \cdot 232 - 15 \cdot 108$

$u = 7, v = -15$

НОД: $\forall \underset{\neq 0}{a}, \underset{\neq 0}{b} \in \mathbb{Z}$, $\exists! c \in \mathbb{N}$: НОД (най-малко общо кр) $[a, b] = c$, ако

1) $a \mid c$ и $b \mid c$
2) $\forall c_1 \in \mathbb{Z}$: $a \mid c_1$ и $b \mid c_1 \Rightarrow c \mid c_1$

ако $a \mid bc$ и $(a, b) = 1 \Rightarrow a \mid c$

(Безу)

$\exists u, v \in \mathbb{Z}$: $a \cdot u + v \cdot b = 1 \mid \cdot c$

$\underbrace{acu}_{a \mid} + \underbrace{vbc}_{a \mid \Rightarrow a \mid} = c$

Нека $\underset{\neq 0}{a}, \underset{\neq 0}{b} \in \mathbb{Z}$ и c е тясно общо кр. Тогава: $c = \frac{t \cdot ab}{(a, b)}$, $t \in \mathbb{Z}$

В частност, ако $a, b \in \mathbb{N}$, то $[a, b] = \frac{ab}{(a, b)}$

ако $(a, b) = 1 \Rightarrow [a, b] = a \cdot b$

$(a, b) = d \Rightarrow d \mid a, d \mid b \Rightarrow \exists a_1, b_1 \in \mathbb{Z}$: $a = d \cdot a_1$, $b = d \cdot b_1$, при това $(a_1, b_1) = 1$

c - общо кратно на a и b

$\Rightarrow a \mid c$ и $b \mid c \Rightarrow \mathbb{Z} \ni \frac{c}{b} = \frac{a \cdot c_1}{db_1} = \frac{d \cdot a_1 \cdot c_1}{db_1}$ и $(a_1, b_1) = 1 \Rightarrow b_1 \mid c_1$

$\Downarrow$

$\exists c_1 \in \mathbb{Z}$: $c = a \cdot c_1$

$\Rightarrow \exists t \in \mathbb{Z}$: $c_1 = tb_1$

$\Rightarrow c = a \cdot c_1 = t \cdot ab_1 = \frac{tab}{d} = \frac{tab}{(a, b)}$ ✓

Зад: $a,b \in \mathbb{Z}$. Да се докаже $(a,b) = (5a+3b, 13a+8b)$

$(a,b) = d$ $\qquad d|k$ и $k|d \Rightarrow d = k$

$(5a+3b, 13a+8b) = k$

$d|a$ и $d|b \Rightarrow d|5a+3b$, $d|13a+8b$ $\Rightarrow d|k$

$k|5a+3b$, $k|13a+8b$ $\Rightarrow$ $k|\underbrace{8(5a+3b)-3(13a+8b)}_{a} \Rightarrow k|a$ $\Rightarrow k|d$

$\Rightarrow k|\underbrace{\quad\quad}_{b} \Rightarrow k|b$

Д.В.:

$(a,b) = 1 \Rightarrow (a+b, a^2+b^2-ab) = 1$ или 3

Прости числа

$p > 1 \in \mathbb{N}$ – просто, ако дел. му са само ± 1 и $\pm p$

ако 1 и не, то е съставно

a е съставно, ако $\exists b,c \in \mathbb{N}: b,c < a$ и $a = b.c$

– $\forall a \in \mathbb{N}$ $\underset{1}{\vee}$ притежава прост дел.

– a – съст. $\Rightarrow a$ притеж. поне 1 прост дел. p: $p \leq \sqrt{a}$

– Простите числа са ∞

– $p \in \mathbb{N}$ – просто $\Leftrightarrow$ ако за a,b: $p|ab \Rightarrow p|a$ или $p|b$

Зад: ДСИ $\forall$ прости числа p, за които $p+4$ и $p+14$ са прости

- $p=2$ ↯
- $p=3 \Rightarrow p+4=7$, $p+14=17$ ✓
- $p=3k+1$ $\quad p+4=3k+5$, $p+14=\underbrace{3k+15}_{3|}$ ↯
- $p=3k+2$ $\quad p+4=\underbrace{3k+6}_{3|}$ ↯

Осн. т-ма на аритм: $\forall a>1 \in \mathbb{N}$ $\exists!$ (с точност до наредба) каноническо представяне

$$a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s}, \quad p_i \neq p_j - \text{прости}, \ \alpha_i \geq 1$$

др. запис:

$a = q_1 \dots q_k$ q_i - не непр. $\neq$ прости с.

$$a = p_1^{\alpha_1} \dots p_s^{\alpha_s}$$
$$b = p_1^{\beta_1} \dots p_s^{\beta_s}$$

$$(a,b) = p_1^{\gamma_1} p_2^{\gamma_2} \dots p_s^{\gamma_s}, \ \gamma_i = \min(\alpha_i, \beta_i)$$

$$[a,b] = p_1^{\delta_1} \dots p_s^{\delta_s}, \ \delta_i = \max(\alpha_i, \beta_i)$$

Числови сравн.

$\forall a, b \in \mathbb{Z}, n \in \mathbb{N}$

$a \equiv b \pmod n \iff n \mid a - b$

a сравнимо с b по mod n

Св-ва:

$\equiv$ - рел. на еквив.

$a \equiv b + c \pmod n$

$\Rightarrow a - c \equiv b \pmod n$

$a_1 \equiv b_1 (n)$

$\vdots$

$a_k \equiv b_k (n)$

$\Rightarrow a_1 + \dots + a_k \equiv b_1 + \dots + b_k (n)$

$a_1 \cdot \dots \cdot a_k \equiv b_1 \cdot \dots \cdot b_k (n)$

$\Rightarrow$ ако $f(x)$ е полином с цели коеф. и $a \equiv b (n) \Rightarrow f(a) \equiv f(b) (n)$

Ако $ka \equiv_k b \pmod n \Rightarrow a \equiv b \pmod{\frac{n}{(k,n)}}$, в частност, ако $(k,n)=1$

$\Rightarrow a \equiv b (n)$

$a \equiv b (n_1)$

$\vdots$

$a \equiv b (n_k)$

$\Rightarrow a \mid b \pmod{[n_1, \dots, n_k]}$, в частност, ако $(n_i, n_j) = 1$, $1 \le i \ne j \le k$

$\Rightarrow a \equiv b \pmod{n_1 \dots n_k}$

Зад: ДСЧ посл. 2 цифри от десетичен запис:

$(207^{19} - 41)^{10} \equiv 5 \pmod{100}$

$207 \equiv 7 \pmod{100}$

$7^2 \equiv 49 \pmod{100}$

$207^3 \equiv 7^3 (100) \equiv 343 (100) \equiv 43 (100)$

$207^4 \equiv 7^4 (100) \equiv 7^3 \cdot 7 (100) \equiv 43 \cdot 7 (100) \equiv 301 (100) \equiv 1 (100)$

$207^{16} = (207^4)^4 \equiv 1^4 (100) \equiv 1 (100)$

$207^{19} = 207^{16} \cdot 207^3 \equiv 1 \cdot 43 (100) \equiv 43 (100)$

$207^{19} - 41 \equiv 43 - 41 \equiv 2 (100)$

$(207^{19} - 41)^{10} \equiv 2^{10} (100) \equiv 24 (100)$

посл. 2 цифри на: участват n седмици

$(\dots((7^7)^7)^7\dots)^7$

$7^7 = 7^4 \cdot 7^3 \equiv 43 (100)$

$7^7 \equiv (7^3)^7 (100) \equiv 7^{21} (100) \equiv (7^4)^5 \cdot 7 (100) \equiv 7 (100)$

$A = \begin{cases} 7 (100) & n = 2k+1 \\ 43 (100) & n = 2k \end{cases}$

Ферма-Ойлер:

Ойлер: $a \in \mathbb{Z}$, $n \in \mathbb{N}$, $(a, n) = 1$

$\Rightarrow a^{\varphi(n)} \equiv 1 (\mathrm{mod}\ n)$, $\varphi(n)$ - ϕ-ция на Ойлер

Ферма: $a \in \mathbb{Z}$, p - просто $\Rightarrow a^p \equiv a (\mathrm{mod}\ p)$

ако $p \nmid a$

$(a, p) = 1 \Rightarrow a^{p-1} \equiv 1 (\mathrm{mod}\ p)$

$\varphi(n)$ - бр. ест. ч. $\leq n$ взаимно пр. с n

Зад: $(85^{70}+19^{32})^{12}$ - ост. при дел. на 21

Ойлер

$(85,21)\equiv 1$ — —

$85\equiv 1(21) \Rightarrow 85^{70}\equiv 1(21)$

$(19,21)\equiv 1$

Ойлер $\Rightarrow 19^{\varphi(21)}\equiv 1(21)$

$\varphi(21)=\varphi(3)\varphi(7)=2\cdot 6=12$ $\quad 3^0\cdot 7^0(3-1)(7-1)$

$\Rightarrow 19^{12}\equiv 1(12) \Rightarrow 19^{24}\equiv 1(12)$

$19\equiv -2(21)$

$(-2)^8\equiv 256\equiv 4(21)$

$\Rightarrow 19^{32}=19^{24}\cdot 19^8\equiv 4(21)$

$85^{70}+19^{32}\equiv 1+4(21)\equiv 5(21)$

$(5,21)=1$

$5^{\varphi(21)}=1(21) \Rightarrow 5^{12}\equiv 1(21)$

посл. 2 цифри на $2^{2026}\equiv r(100)$

$(2,100)=2$ ~~Ойлер~~

$100=4\cdot 25, \quad (4,25)=1$

$2^{2026}\equiv r(4)$

$2^{2026}\equiv r(25)$ $\Rightarrow 2^{2026}\equiv r(4\cdot 25)$

$2^{2026}\equiv 0(4)\equiv 4(4)\equiv \overset{64(4)}{---}\equiv 96(4)$

$(2,25)=1 \overset{\text{Ойлер}}{\Rightarrow} 2^{\varphi(25)}\equiv 1(25)$

$\varphi(25) = 5.4 = 20$

$\Rightarrow 2^{20} \equiv 1(25) \Leftrightarrow 2^{2026} \equiv 1(25)$

$2^6 \equiv 64(25) \equiv 14(25) \equiv 39(25) \equiv 89(25)$

$\Rightarrow 2^{2026} \equiv 64(100)$

Д.У. a-четно, а не се дели на 10. ДСН посл. 2 цифри на a^{20}
—"— 3 ц. на a^{100}

Зад: ДСД, че $2^{11 \cdot 31} \equiv 2 \pmod{11 \cdot 31}$

Ферма $2^{11 \cdot 31} \equiv 2(11)$, $2^{11 \cdot 31} \equiv 2(31)$, $(11, 31) = 1$

Ферма $\Rightarrow 2^{11} \equiv 2(11)$

$\Rightarrow 2^{11 \cdot 31} \equiv 2^{31}(11)$

$(2, 11) = 1 \Rightarrow 2^{10} \equiv 1(11) \Rightarrow 2^{30} \equiv 1(11) \Leftrightarrow 2^{31} \equiv 2(11)$ ✓

$\Rightarrow 2^{31} \equiv 2(31)$

$2^{11 \cdot 31} \equiv 2^{11}(31)$

$2^5 - 32 \equiv 1(31) \Rightarrow 2^{11} \equiv 2^{10} \cdot 2 \equiv 2(31)$ ✓

Д.У. $2^{70} + 3^{70}$ се дели на 13

23.04. F-поле

$\operatorname{char} F = 0$ или p-просто

↳ ред на ел. 1 в $(F, +)$

ако $|1| = \infty$ в $(F, +) \Rightarrow \operatorname{char} F = 0$

ако $|1| < \infty$ в $(F, +) \Rightarrow |1| = p$ (p просто) и $\operatorname{char} F = p$

$\Rightarrow \forall a \in F: pa = \underbrace{a+a+\dots+a}_{p\text{-пъти}} = 0$

$x, y \in F \Rightarrow (x+y)^p = x^p + y^p$ (F-поле, $\mathrm{char} F = p$, Нют. бином)

$x^p + \underbrace{\binom{p}{1}x^{p-1}y + \dots + \binom{p}{p-1}xy^{p-1}}_{p\,|} + y^p$

$n \in \mathbb{N} \Rightarrow (x+y)^{p^n} = x^{p^n} + y^{p^n}$

F - крайно поле $\Rightarrow \mathrm{char} F \neq 0$

F - безкрайно поле $\Rightarrow \mathrm{char} F = 0$ или p

Пр за $\forall$ поле с кр. характ $(=p)$: $F = \left\{ \frac{f(x)}{g(x)} \mid f, g \in \mathbb{Z}_p[x], g \neq 0 \right\}$

Важна зад: Нека F е кр. поле с $\mathrm{char} F = p$, тогава $|F| = p^n$, $n \in \mathbb{N}$

K - просто поле = няма собствени подполета $(\neq K)$

F - поле $\Leftrightarrow$ то притежава единствено просто подполе $K = \bigcap_{H_i \leq F} H_i$

ако $\mathrm{char} F = p \Rightarrow \exists! K$ - просто подполе: $K \cong \mathbb{Z}_p$

$\mathrm{char} F = 0 \Rightarrow \exists! K$ - — // — : $K \cong \mathbb{Q}$

$\Rightarrow \exists K_0 < K : K_0$ - просто $\Rightarrow K_0 \cong \mathbb{Z}_p$: K е лин. пр-во над $\mathbb{Z}_p$

$\Rightarrow \exists a_1, a_2, \dots, a_n \in K$ - базис на K над $\mathbb{Z}_p$

$\forall a \in K: a = \sum_{i=1}^{n} \lambda_i a_i,\ \lambda_i \in \mathbb{Z}_p \Rightarrow |K| = p^n$

Зад:

Да се конструира поле с 4 ел. и да се док, че $\forall a$ такива полета са изом.

$|F|=4 \Rightarrow \operatorname{char} F=2 \Rightarrow 2a=0$, $\forall a \in F \Leftrightarrow a=-a$ (1)

$|F^*|=3 \Rightarrow F^* \cong \mathbb{C}_3 \Rightarrow F^*=\{1,g,g^2\}=\langle g \rangle \Rightarrow F=\{0,1,g,g^2\}$

↓ мултипл. гр.

$g^3=1 \Rightarrow 0=g^3-1=(g-1)(g^2+g+1)$

(*)!! $(g+1)$

1сл. $g=1$

2сл. $g \neq 1 \Rightarrow g^2+g+1=0$

$1+g=g^2$

$1+g^2=g$

$g+g^2=1$

+	0	1	g	g^2
0	0	1	g	g^2
1	1	0	g^2	g
g	g	g^2	0	1
g^2	g^2	g	1	0

·	0	1	g	g^2
0	0	0	0	0
1	0	1	g	g^2
g	0	g	g^2	1
g^2	0	g^2	1	g

R-пръстен

$\emptyset \neq I \subseteq R$

$\forall a,b \in I, r \in R$

1) $a-b \in I \Rightarrow (I,+) < (R,+)$

2) $a \in I$ - десен идеал
$ra \in I$ - ляв идеал

ако и двете са изпълнени $\Rightarrow I$ е (двустранен) идеал

$I \trianglelefteq R$

$$I = \left\{ \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ & 0 & & \end{pmatrix}_{n \times n} \mid a_{ij} \in F \right\}$$

$R = M_n(F)$, $(I, +) < (R, +)$

$\forall A \in I, B \in R \Rightarrow AB \in I \Rightarrow I$ е десен идеал
$BA \notin I$

ч.ч. $R = \{a, b, c, d\}$. ОСН идеали и подгр. на R

+	a	b	c	d
a	a	b	c	d
b	b	c	d	a
c	c	d	a	b
d	d	a	b	c

·	a	b	c	d
a	a	a	a	a
b	a	b	c	d
c	a	c	a	c
d	a	d	c	b

	идеали	подгр.
тривиални	$\{a\}, R$	$\{a\}, R$
нетрив.	$\{a, b\}$ ($b+b=a$), $\{a, c\}$, $\{a, d\}$	$\{a, b\}$, $\{a, c\}$, $\{a, d\}$

$|(R, +)| = 4 \overset{\text{Лагранж}}{\Rightarrow} \forall H < (R, +) : |H| = 1, 2$, или 4

$\Rightarrow$ няма да разгл. 3-елементни подгр-ва на R (не могат да образ. гр)

ОР:

Зад:

+	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	a	b
d	d	c	b	a

·	a	b	c	d
a	a	a	a	a
b	a	b	c	d
c	a	c	c	a
d	a	d	a	d

R -пр. с 1

$\phi \neq X \subseteq R$

$(X) = \{x_1 r_1 + x_2 r_2 + \dots + x_n r_n \mid x_i \in X, r_i \in R\}$

док. се, че $(X) \trianglelefteq R$

↳ идеал, породен от множ. X

Ако $X = \{a\} \Rightarrow (a) = \{ar \mid r \in R\} \trianglelefteq R$

↳ главен идеал, породен от ел. a

В $\mathbb{Z}$ ∀ идеал е главен

$I \trianglelefteq \mathbb{Z} \Rightarrow (I,+) < (\mathbb{Z},+)$

↳ ∞ цикл. гр.

$\Rightarrow (I,+)$ е ∞ цикл.

$\Rightarrow (I,+) = k\mathbb{Z}, k \in \mathbb{N} \cup \{0\}$

очевидно ел. на $k\mathbb{Z}$ изчерпват на цялото с ∀ ел. k

$\Rightarrow (k) = k\mathbb{Z} \trianglelefteq \mathbb{Z}$

В $\mathbb{Z}_n$ $\forall$ идеал е главен.

R – пр. с 1, $I \trianglelefteq R$

ОСФ, че $I = R \Leftrightarrow I$ съдържа обратим ел.

$\Rightarrow$ $I = R \Rightarrow 1 \in I$ (обратим)

$\Leftarrow$ $\exists a \in I : \exists a^{-1} \in R$

$\underset{\in I}{a}\underset{\in R}{a^{-1}} = 1 \in I \Rightarrow (1) = R$

$\forall r \in R : r = r \cdot \underset{\in I}{1} \in I \Rightarrow I = R$

$\Rightarrow$ В поле няма собств. идеали

Зад:

$I \trianglelefteq R \Rightarrow (I,+) \leq (R,+) \Rightarrow R/I$ – фактор гр.

$(R,+)/(I,+)$

$R/I = \{ I, a+I, a^2+I, \dots \}$

Дефиниране умнож. в R/I: $(a+I)(b+I) = ab+I$

$\bar{a}\bar{b} = \overline{ab}$

– коректност

– R/I е фактор гр. за $+_n$ – абелева

– асоц. на $\cdot_n$ – наследена от R

– дистр. закон – – ν – –

$\Rightarrow R/I$ – пръстен (фактор пръстен)

$I \trianglelefteq R$:

$\bar{\varphi}: R \to R/I$

$\bar{\varphi}(a) = a + I = \bar{a}$

$\bar{\varphi}$ е ест. хом. на пръстена

$$\ker \bar{\varphi} = \{ a \in R \mid \bar{\varphi}(a) = \underset{\text{ч}}{\bar{0}} \} = I$$

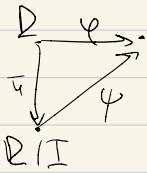
$\Rightarrow$ $\forall$ идеал е ядро на хом.

$\varphi: R \to R'$ - хом.

$\ker \varphi = \{ a \in R \mid \varphi(a) = 0_{R'} \} \trianglelefteq R$

$\operatorname{Im} \varphi = \{ b \in R' \mid \exists a \in R : \varphi(a) = b \} \leq R'$

Т-ма за хом.: $\varphi: R \to R'$ - хом. и $I = \ker \varphi$
$\Rightarrow R/I \cong \operatorname{Im} \varphi$ и $I \trianglelefteq R$



4.7. p-просто

$R = \{ \frac{m}{n} \mid m,n \in \mathbb{Z}, (m,n) = 1, p \nmid n \}$, $M = \{ \frac{m}{n} \in R \mid p \mid m \}$

а) Да се докаже, че R е пр. с 1 и $M \trianglelefteq R$

б) Да се намерят обратимите ел. на R и да се докаже, че всеки собствен идеал на R се съдържа в M.

в) $R/M \cong \mathbb{Z}_p$

$\varphi \neq R \subset Q$

$R \leq Q$

1) $\frac{m_1}{n_1} - \frac{m_2}{n_2} = \underbrace{\frac{m_1 n_2 - m_2 n_1}{n_1 n_2}}_{\in \mathbb{Z}} \in R$, ако $(m_1 n_2 - m_2 n_1, n_1 n_2) = d > 1 \Rightarrow$ съкращ. на d и накрая дроб. е несъкратима

$p \nmid n_1, p \mid n_2 \Rightarrow p \nmid n_1 n_2$

$\Rightarrow (R, +) < (Q, +)$

2) $\frac{m_1}{n_1} \cdot \frac{m_2}{n_2} \in R$ – както в 1) $\Rightarrow R < Q$

3) $1 = \frac{1}{1} \in R$ $(p \nmid 1)$

4) R е комут. пр. с 1 (Q е поле)

$M \lhd R$ (макс от т-ма за хом.)

$\frac{a_1}{b_1}, \frac{a_2}{b_2} \in M, \frac{m}{n} \in R$

1) $\frac{a_1}{b_1} - \frac{a_2}{b_2} = \frac{a_1 b_2 - a_2 b_1}{b_1 b_2} \in M$, $p \mid a_1, p \mid a_2 \Rightarrow p \mid a_1 b_2 - a_2 b_2$

2) $\frac{a_1}{b_1} \cdot \frac{m}{n} = \frac{m}{n} \cdot \frac{a_1}{b_1} = \frac{m a_1}{n b_1} \in M$, $p \mid a_1 \Rightarrow p \mid m a_1$

$\Rightarrow M \lhd R$

а) $\frac{a}{b} \in \mathbb{R}$ е обратим, ако $\frac{b}{a} \in \mathbb{R}$

$\frac{b}{a} \in \mathbb{R} \Leftrightarrow p \nmid b, a$

$\Leftrightarrow M$ е множ. от $\forall$ необр. и $\mathbb{R}/M$ - множ. на обратимите

$\forall I \trianglelefteq \mathbb{R}$ съдържа само необр. ел. $\Rightarrow I \subseteq M \trianglelefteq_{max} \mathbb{R}$

б) $\varphi : \mathbb{R} \to \mathbb{Z}_p$

$\frac{a_1}{b_1} + M = \frac{a_2}{b_2} + M \Leftrightarrow \frac{a_1}{b_1} - \frac{a_2}{b_2} \in M \Leftrightarrow \frac{a_1 b_2 - a_2 b_1}{b_1 b_2} \in M$

$\Leftrightarrow p \mid a_1 b_2 - a_2 b_1 \Leftrightarrow a_1 b_2 \equiv a_2 b_1 \pmod p$ $\Big|$. b_1^{-1}, b_2^{-1} на $p \nmid b_1, p \nmid b_2$, $\exists b_1^{-1}, b_2^{-1} \in \mathbb{Z}_p$

$\Leftrightarrow a_1 b_1^{-1} \equiv a_2 b_2^{-1} \pmod p$

$\varphi\left(\frac{a}{b}\right) = \bar{a}\,\bar{b}^{-1} = \overline{ab^{-1}}$

φ е хом. на пресечки

1) $\varphi\left(\frac{a_1}{b_1} + \frac{a_2}{b_2}\right) = \varphi\left(\frac{a_1 b_2 + a_2 b_1}{b_1 b_2}\right) = \overline{(a_1 b_2 + a_2 b_1)}\ \overline{(b_1 b_2)^{-1}} =$

$= (\bar{a}_1 \bar{b}_2 + \bar{a}_2 \bar{b}_1)(\bar{b}_1^{-1} \cdot \bar{b}_2^{-1}) = \bar{a}_1 \bar{b}_1^{-1} + \bar{a}_2 \bar{b}_2^{-1} = \varphi\left(\frac{a_1}{b_1}\right) + \varphi\left(\frac{a_2}{b_2}\right)$

2) $\varphi\left(\frac{a_1}{b_1}\frac{a_2}{b_2}\right) = \overline{(a_1 a_2)}\ \overline{(b_1 b_2)^{-1}} = \bar{a}_1 \bar{b}_1^{-1} \bar{a}_2 \bar{b}_2^{-1} = \varphi\left(\frac{a_1}{b_1}\right) \cdot \varphi\left(\frac{a_2}{b_2}\right)$ ✓

$\Rightarrow \varphi$ е хом

3) φ е сюрекция

$-\operatorname{Im}\varphi \leq \mathbb{Z}_p$

$-\forall \bar{m} \in \mathbb{Z}_p \quad \exists \frac{m}{1} \in \mathbb{R}: \varphi\left(\frac{m}{1}\right) = \bar{m} \cdot \bar{1}^{-1} = \bar{m} \Rightarrow \mathbb{Z}_p \leq \operatorname{Im}\varphi$

$\Rightarrow \mathbb{Z}_p = \operatorname{Im}\varphi$

4) $\operatorname{Ker}\varphi = \{ \frac{a}{b} \in \mathbb{R} \mid \varphi(\frac{a}{b}) = \bar{0} \} = \{ \frac{a}{b} \in \mathbb{R} \mid \bar{a}\bar{b}^{-1} = 0 \} = M$

$\Leftrightarrow a = \bar{0}$

$\Leftrightarrow p \mid a$

Т-ма за хом. $\Rightarrow$ $R/M \cong \mathbb{Z}_p$

Зад: $I = (2+\sqrt{-1}) \lhd \mathbb{Z}[\sqrt{-1}]$, където $\mathbb{Z}[\sqrt{-1}] = \{a+b\sqrt{-1} \mid a, b \in \mathbb{Z}\}$ → пръстен на целите Гаусови числа

Да се пр., че $I = J = \{a+b\sqrt{-1} \mid 5 \mid b+2a\}$ и $\mathbb{Z}[\sqrt{-1}]/I \cong \mathbb{Z}_5$

1) $I \subseteq J$

$(2+\sqrt{-1})(a+b\sqrt{-1}) = \underbrace{2a-b}_{A} + \underbrace{(2b+a)}_{B}\sqrt{-1} \in I$

? $5 \mid B+2A \Leftrightarrow 2b+a+4a-2b = \underbrace{5a}_{5\mid}$, $a \in \mathbb{Z}$

$\Rightarrow I \subseteq J$

2) $\mathcal{J} \subseteq \mathcal{I}$

$A + B\sqrt{-1} \in \mathcal{J} \Leftrightarrow 5 \mid B + 2A$

$\exists a, b \in \mathbb{Z}$:

$$\begin{cases} A = 2a - b \mid \cdot 2 \\ B = 2b + a \end{cases} +$$

$2A + B = 4a - 2b + 2b + a = 5a \Leftrightarrow a = \frac{2A+B}{5} \in \mathbb{Z}$ (от деф)

$$\begin{cases} A = 2a - b \\ B = 2b + a \mid \cdot(-2) \end{cases} +$$

$A - 2B = 2a - b - 4b - 2a^2 - 5b$

$b = \frac{2B - A}{5} \stackrel{?}{\in} \mathbb{Z}$

$2A + B \equiv 0 \pmod 5$ по усл.

$2A \equiv -B \pmod 5 \mid \cdot(-2)$

$-4A \equiv 2B \pmod 5$

$-4 \equiv 1 \pmod 5$

$A \equiv 2B \pmod 5 \Leftrightarrow 5 \mid 2B - A$

$\Rightarrow b = \frac{2B - A}{5} \in \mathbb{Z}$

$A + B\sqrt{-1} = 2a - b + (2b + a)\sqrt{-1} = (2 + \sqrt{-1})(a + b\sqrt{-1}) \in \mathcal{I} \Rightarrow \mathcal{J} \subseteq \mathcal{I}$ ∨

$\Rightarrow \mathcal{I} = \mathcal{J}$

$\mathbb{Z}[\sqrt{-1}]/I$

$a_1+b_1\sqrt{-1} \overset{+I}{=} a_2+b_2\sqrt{-1}+I \iff a_1-a_2+(b_1-b_2)\sqrt{-1} \in I$

$\iff 5 \mid b_1-b_2+2(a_1-a_2)$

$\iff b_1+2a_1 \equiv b_2+2a_2 \pmod 5$

$\varphi: \mathbb{Z}[\sqrt{-1}] \to \mathbb{Z}_5$

$\varphi(a+b\sqrt{-1}) = \overline{b+2a}$

φ е хом:

1) $\varphi((a_1+b_1\sqrt{-1})+(a_2+b_2\sqrt{-1})) = \overline{b_1+b_2+2(a_1+a_2)} = \overline{b_1+2a_1}+\overline{b_2+2a_2}$

$= \varphi(a_1+b_1\sqrt{-1})+\varphi(a_2+b_2\sqrt{-1})$

2) $\varphi((a_1+b_1\sqrt{-1})(a_2+b_2\sqrt{-1})) = \varphi(a_1a_2-b_1b_2+(a_1b_2+a_2b_1)\sqrt{-1}) =$

$= \overline{a_1b_2+a_2b_1+2(a_1a_2-b_1b_2)}$

$\varphi(a_1+b_1\sqrt{-1})\varphi(a_2+b_2\sqrt{-1}) = \overline{(b_1+2a_1)}\,\overline{(b_2+2a_2)} = \overline{4a_1a_2+b_1b_2+2(a_1b_2+a_2b_1)}$

$= \overline{2}\,\varphi((a_1+b_1\sqrt{-1})(a_2+b_2\sqrt{-1}))$

$\psi: \mathbb{Z}[\sqrt{-1}] \to \mathbb{Z}_5$

$\psi(a+b\sqrt{-1}) = \bar{k}\varphi(a+b\sqrt{-1})$

$\psi = \bar{k}\varphi$

+търсим $\bar{k} \in \mathbb{Z}_5$ т.ке ψ е хом. на пръстени

1) $\psi((a_1+b_1\sqrt{-1})+(a_2+b_2\sqrt{-1})) = \bar{k}\,\varphi((\dots)+(\dots)) = \bar{k}\,\varphi(\dots)+\bar{k}\,\varphi(\dots)$

$\psi(\dots) \quad \varphi(\dots)$ ✓

2) $\varphi((a_1+b_1\sqrt{-1})(a_2+b_2\sqrt{-1})) = \bar{u}\,\varphi(\dots)(\dots)$

u

$\varphi(\dots)\varphi(\dots) = \bar{u}\varphi(\dots)\,\bar{u}\varphi(\dots) = \bar{u}^2\varphi(\dots)\varphi(\dots) = 2\bar{u}^2\ \varphi(a_1+b_1\sqrt{-1})\cdot\varphi(a_2+b_2\sqrt{-1})$

$\varphi((\dots)(\dots)) = \varphi(\dots)\varphi(\dots) \iff \bar{u}\varphi((\dots)(\dots)) = \bar{2}\,\bar{u}^2\varphi((\dots)(\dots))$

$\iff 2\bar{u} = \bar{1}$ in $\mathbb{Z}_8$

$\iff \bar{u} = 3$

$\Rightarrow \varphi(a+b\sqrt{-1}) = \overline{a+3b}$

$\Rightarrow \varphi$ ex.

3) $\mathrm{Im}\,\varphi = \mathbb{Z}_5$

$\forall\, \bar{c}\in\mathbb{Z}_5:\ c+0\sqrt{-1}\in\mathbb{Z}[\sqrt{-1}]:\ \varphi(c+0\cdot\sqrt{-1}) = \bar{c}$

4) $\ker\varphi = \{a+b\sqrt{-1} \mid \varphi(a+b\sqrt{-1}) = \bar{0}\} = I$

$\overline{a+3b} = \bar{0}$

$a \equiv -3b\ (5)\ /\cdot 2$

$2a \equiv -b\ (5) \iff 5 \mid 2a+b$

$\Rightarrow \mathbb{Z}[\sqrt{-1}]/I \cong \mathbb{Z}_5$

1) $f \mid f$

2) $f \mid g$ и $g \mid f \Rightarrow f = \alpha g, \alpha \in F^*$

3) $f \mid g$ и $g \mid h \Rightarrow f \mid h$

4) $f \mid g$ и $f \mid h \Rightarrow f \mid (g \pm h)$

5) $f \mid g$ и $f \mid g + h \Rightarrow f \mid h$

6) $f \mid g$, $\alpha, \beta \in F^* \Rightarrow \alpha f \mid \beta g$

Т-ма за дел. с ост: За $\forall f, g \in F[x]$ $\exists! q, r \in F[x]$: $f = g \cdot q + r$, $\deg r < \deg g$ (ако F е поле; q – частно, r – остатък)

ако F е обл. на цялост: g има обратим коеф.

НОД на f и g: $d = (f, g) \in F[x]$

1) $d \mid f$ и $d \mid g$

2) $\forall d_1 \mid f$ и $d_1 \mid g$: $d_1 \mid d$

ако $d \mid d_1 \Rightarrow d = \alpha d_1, \alpha \in F^*$

3) d е със старши коеф. 1

Евклид

$f = gq_1 + r_1$ $\quad \deg r_1 < \deg g$

ако $r_1 \neq 0$:

$g = r_1 q_2 + r_2$ $\quad \deg r_2 < \deg q_1$

$\vdots$

$r_{k-2} = r_{k-1} q_k + r_k$

ако $r_k \neq 0$ $\quad \deg r_k < \deg r_{k-1}$

$r_{k-1} = r_k q_{k+1} + 0$

$\Rightarrow (f, g) = r_k$

Безу: $\forall f,g \in F[x] \ \exists u,v \in F[x] : uf+vg=(f,g)$ и $(u,v)=1$

Заг: ДСН НОД на $f,g \in F[x]$ такъв че $u,v \in F[x] : uf+vg=d$

$f = x^5-3x^3+2x^2+x-2$ | $g = x^3-3x+1$, $F=\mathbb{Q}$

$-\ x^5-3x^3+x^2$ | $q_1 = x^2$

$r_1 = x^2+x-2$ | $\deg r_1 < \deg g$

$g = x^3-3x+1$ | $r_1 = x^2+x-2$

x^3+x^2-2x | $q_2 = x-1$

$-x^2-x+1$

$-x^2-x+2$

$r_2 = -1$

$\Rightarrow (f,g) = -r_2 = 1$

$f = g \cdot q_1 + r_1$

$g = r_1 \cdot q_2 + r_2$

$r_2 = g - r_1 q_2 = g - (f - g \cdot q_1) q_2 = -q_2 f + (1+q_1 q_2) g$

$\Rightarrow u = -(-q_2) = q_2 = x-1$

$v = -(1+q_1 q_2) = -x^3+x^2-1$

$f = x^5-2x^3+x^2-3x+1$ | $g = x^3-3x+1$, $F=\mathbb{Q}$

$-\ x^5-3x^4+x^2$ | $q_1 = x^2+1$

x^3-3x+1

$-\ x^3-3x+1$

0

$\Rightarrow (f,g) = g$

$f = g \cdot q_1$

$u \cdot f + v \cdot g = g$

$u = 0$

$v = 1$

$f = x^4 - 2x^3 + 2x - 4$ | $g = x^3 - 2x^2 + 4x - 8$, $F = \mathbb{Q}$

$- x^4 - 2x^3 + 4x^2 - 8x$ | $q_1 = x$

$r_1 = -4x^2 + 10x - 4$

$g = x^3 - 2x^2 + 4x - 8$ | $r_1 = -4x^2 + 10x - 4$

$- x^3 - \frac{5}{2}x^2 + x$ | $q_2 = -\frac{1}{4}x - \frac{1}{8}$

$\frac{1}{2}x^2 + 3x - 8$

$- \frac{1}{2}x^2 - \frac{5}{4}x + \frac{1}{2}$

$r_2 = \frac{17}{4}x - \frac{17}{2}$

$r_1 = -4x^2 + 10x - 4$ | $r_2 = \frac{17}{4}x - \frac{17}{2}$

$- -4x^2 + 8x$ | $q_3 = -\frac{16}{17}x + \frac{8}{17}$

$-2x - 4$

$2x - 4$

0

$(f, g) = \frac{4}{17} r_2 = x - 2$

$f = g \cdot q_1 + r_1$

$g = r_1 q_2 + r_2$

$r_2 = g - r_1 q_2 = g - (f - g q_1) q_2 = -q_2 f + (1 + q_1 q_2) g$

$u = \frac{4}{17}(-q_2) = \frac{1}{17}x + \frac{1}{34}$

$v = \frac{4}{17}(1 + q_1 q_2) = -\frac{1}{17}x^2 - \frac{1}{34}x + \frac{4}{17}$

$f = x^4 + x^3 - \bar{4}x^2 + x + \bar{1}$ | $g = x^4 - x^3 + x - \bar{3}$, $F = \mathbb{Z}_7$

$- x^4 - x^3 + x - \bar{3}$ | $q_1 = \bar{1}$

$r_1 = \bar{2}x^3 - \bar{4}x^2 + \bar{4}$

$g = x^4 - x^3 + x - \bar{3}$ | $r_1 = \bar{2}x^3 - \bar{4}x^2 + \bar{4}$

$- x^4 - \bar{2}x^3 + \bar{2}x$ | $q_2 = \bar{4}x + \bar{4}$

$x^3 - x - \bar{3}$

$- x^3 - \bar{2}x^2 + \bar{2}$

$r_2 = 2x^2 - x - 5$

$r_1 = \bar{2}x^3 - \bar{4}x^2 + \bar{4}$

$-\bar{2}x^3 - x^2 - \bar{5}x$

$-\bar{3}x^2 + \bar{5}x + \bar{4}$

$-\bar{4}x^2 - \bar{2}x - \bar{10} = -\bar{3}$

0

$r_2 = \bar{2}x^2 - x - \bar{5}$

$q_3 = x + \bar{2}$

$(f,g) = \bar{4}r_2 = x^2 - \bar{4}x + \bar{1} = x^2 + \bar{3}x + \bar{1}$

$uf + gv = \bar{4}r_2$

$r_2 = -q_2 f + (\bar{1} + q_1 q_2) g$

$u = \bar{4}(-q_2) = \bar{5}x + \bar{5}$

$v = \bar{4}(\bar{1} + q_1 q_2) = \bar{2}x + \bar{6} = \bar{2}x - \bar{1}$

$f = \bar{5}x^4 + \bar{3}x^3 + \bar{5}x^2 + \bar{6}x + \bar{3}$

$\bar{5}x^4 + \bar{2}x^3 \quad -\bar{4}x - \bar{3}$

$r_1 = x^3 + \bar{5}x^2 + \bar{10}x + \bar{6}$

$g = x^4 + \bar{7}x^3 + \bar{8}x + \bar{6}$

$q_1 = \bar{5}$

$F = \mathbb{Z}_{11}$

$g = x^4 + \bar{5}x^3 + \bar{10}x^2 + \bar{6}x$

$r_2 = \bar{2}x^3 + x^2 + \bar{2}x + \bar{6}$

$-10x^2$

$-\bar{2}x^3 + \bar{10}x^2 - \bar{2}x + \bar{1}$

$r_2 = \bar{2}x^2 + \bar{4}x + \bar{5}$

$r_1 = x^3 + \bar{5}x^2 + \bar{10}x + \bar{6}$

$q_2 = x + \bar{2}$

$r_1 = x^3 + \bar{2}x^2 - \bar{3}x$

$3x^2 + \bar{2}x + \bar{6}$

$3x^2 + \bar{6}x + \bar{2}$

$r_2 = \bar{2}x^2 + \bar{4}x + \bar{5}$

$q_3 = \bar{6}x + \bar{7}$

$r_3 = \bar{4}x + \bar{4}$

$r_2 = \bar{2}x^2 + \bar{4}x + \bar{5}$ | $r_3 = \bar{4}x + \bar{4}$

$-\bar{2}x^2 - \bar{2}x$ | $q_4 = \bar{5}x + \bar{4}$

$\bar{6}x + \bar{5}$

$-\bar{6}x + \bar{5}$

$\bar{0}$

$(f, g) = \bar{8} r_3 = x - \bar{1}$

т.е. $f + U \cdot g = x - \bar{1}$

$f = g \cdot q_1 + r_1$

$g = r_1 q_2 + r_2$

$r_1 = r_2 q_3 + r_3$

$r_3 = r_1 - r_2 q_3 = r_1 - (g - r_1 q_2) \cdot q_3 = (-q_3) g + (\bar{1} + q_2 q_3) r_1 = (-q_3) g + (\bar{1} + q_2 q_3)(f - g q_1)$

$= (\bar{1} + q_2 q_3) f + (-q_1 - q_3 - q_1 q_2 q_3) g$

$u = \bar{8}(\bar{1} + q_2 q_3) \qquad v = \bar{3}(q_1 + q_3 + q_1 q_2 q_3)$

$f \in F[x]$, $F \subset K$ — поле, разширение на F-поле

ако $\exists \alpha \in K: f(\alpha) = 0 \to \alpha$ е корен на $f(x)$

$f \in F[x]$ е разложим (оставени числа), ако $\exists g, h \in F[x]: f = g \cdot h$, то $\deg g < \deg f$ и $\deg h < \deg f$

В противен сл. е неразложим: $f = g \cdot h \Rightarrow \deg g = 0, \deg h = f$ или обратното

(простите числа)

Не разложимост над $\mathbb{C}$ – на полиноми от $1^{ва}$ степен (лин. множ.)
над $\mathbb{R}$ – полиноми от $1^{ва}$ и такива от $2^{ра}$ ст. с $D<0$
над $\mathbb{Q}$ – от произволно висока ст.

Кр. на Айзенщайн: Нека $f \in \mathbb{Z}[x]$, $f = a_0x^n + a_1x^{n-1} + \dots + a_n$

p – просто такова, че:

1) $p \nmid a_0$
2) $p \mid a_1, \dots, a_n$
3) $p^2 \nmid a_n$

$\Rightarrow f$ е неразложим над $\mathbb{Q}$

$\Rightarrow f_{(x+1)} = x^n + p \Rightarrow$ неразл. над $\mathbb{Q}$

За се разложи над $\mathbb{R}, \mathbb{C}, \mathbb{Q}$

$f(x) = x^4+16 = x^4+8x^2+16-8x^2 = (x^2+4)^2 - 8x^2 = (x^2+2\sqrt{2}x+4)(x^2-2\sqrt{2}x+4)$

$x_{1,2} = -\sqrt{2} \pm \sqrt{2}i$

$\mathbb{Q}$ $\quad$ $D<0$ $\quad$ $\mathbb{R}$

$= (x-\sqrt{2}(1+i))(x-\sqrt{2}(1-i))(x+\sqrt{2}(1+i))(x+\sqrt{2}(1-i))$

$\mathbb{C}$

$\mathbb{R}$ и $\mathbb{C}$

$f(x) = x^4+x^3-4x^2-5x-5 = x^4+x^3+x^2-5x^2-5x-5 = x^2(x^2+x+1)-5(x^2+x+1) = (x^2-5)(x^2+x+1)$ $\quad \mathbb{Q}$

$= (x-\sqrt{5})(x+\sqrt{5})(x^2+x+1) = (x-\sqrt{5})(x+\sqrt{5})\left(x+\frac{1+\sqrt{3}i}{2}\right)\left(x+\frac{1-\sqrt{3}i}{2}\right)$

$\mathbb{R}$ $\quad$ $\mathbb{C}$

Да се докаже, че f е неразл. над $\mathbb{Q}$

$f(x) = x^4+px+q$, p, q – прости и $(p,q) \neq (3,2)$

проверка за корени

кандидати: $\pm 1, \pm q$ (дел. на своб. коеф)

$f(1) = 1+p+q > 0$ ✗

$f(-1) = 1-p+q = 0 \Leftrightarrow p = q+1 \Leftrightarrow (p,q) = (3,2)$ – забранено по усл. ✗

$f(q) = q^4 + pq + q > 0$ ✗

$f(-q) = q^4 - pq + q = q(q^3 - p + 1) = 0 \Leftrightarrow q^3 = p - 1$ (иначе $q^3 = 1$ ↯)

$\Rightarrow p \neq 2 \Rightarrow p$ – нечетно

$\Rightarrow p-1$ – четно $\Rightarrow q = 2$, но $2^3 = 9 - 1$ ↯ $p \neq 9$

$\Rightarrow f$ няма корени

? $\exists g, h \in \mathbb{Z}[x]: f = gh$ и $\deg g = \deg h = 2$

$f = x^4 + px + q = (x^2 + ax + b)(x^2 + cx + d)$ – метод на неопр. коеф

$x^3 \to 1)$	$0 = a + c$	1)	$a = -c$
$x^2 \to 2)$	$0 = b + d + ac$	2)	$b + d - c^2 = 0$
$x \to 3)$	$p = ad + bc$	3)	$(b-d) \cdot c = p$
$x^0 \to 4)$	$q = bd$	4)	$bd = q$

1 сл. $b = 1, d = q$

2) $\Rightarrow c^2 = q + 1$

3) $\Rightarrow (1-q) \cdot c = p$

$(1-q) \mid p \Rightarrow 1 - q = \pm 1, \pm p \Rightarrow -1$ или $-p$

$1 - q = -1 \Rightarrow q = 2 \overset{2)}{\Rightarrow} c^2 = 3$ ↯

$1 - q = -p \Rightarrow q = 1 + p \Rightarrow p = 2, q = 3 \Rightarrow c^2 = 4 \Rightarrow c = \pm 2$ ↯

$c = 2 \Rightarrow 1 - q = 1$ ↯

$c = -2 \Rightarrow 1 - q = -1$ ↯

2 сл. $b = -1, d = -q$

$\overset{3)}{\Rightarrow} c^2 = -1 - q < 0$ ↯

$\Rightarrow f$ е неразложимо

14.05.

9.бан

$f = x^4 + 6x^2 + 8x + 9$ ДСД, че f е неразложим над $\mathbb{Q}$

проверка за рационални корени

кандидати: $\pm 1, \pm 3, \pm 9$

$f(1), f(3), f(9) > 0$

	1	0	6	8	9
-1	1	-1	7	1	8
-3	1	-3	15		
-9	1	-9	87		

$\Rightarrow$ няма корени

$\exists ? g, h \in \mathbb{Z}[x]: f = gh,\ \deg g = 2 = \deg h$ неразложими

$f = x^4 + 6x^2 + 8x + 9 = (x^2 + ax + b)(x^2 + cx + d)$

$x^3 \to$	1)	$a + c = 0$	$a = -c$
$x^2 \to$	2)	$b + d + ac = 6$	$b + d - c^2 = 6$
$x^1 \to$	3)	$ad + bc = 8$	$(b - d)c = 8$
$x^0 \to$	4	$bd = 9$	$bd = 9$

$\Rightarrow$

1 сл: $b = d = 3$

	1)	$a = -c$
	2)	$6 - c^2 = 6 \Leftrightarrow c^2 = 0 \Leftrightarrow c = a = 0$
$\Rightarrow$	3)	$0 = 8$ ↯

2 сл. $b = d = -3$

3) $0 = 8$ ↯

за $b=1$ $d=9$ ($b=9$, $d=1$ е същото!)

$a=-c$

$10-c^2=6 \Rightarrow c^2=4 \Rightarrow c=\pm 2$

$-8 \cdot c=8 \Rightarrow c=-1$ ↓

за $b=-1$, $d=-9$ ($=$ $b=-9$, $d=-1$)

$\Rightarrow$ 2) $c^2=-16$ ↓

$\Rightarrow$ f е неразложим над $\mathbb{Q}$

З.Т. $\alpha, \beta \in F$-поле, $\alpha \neq 0$. ДСД, че $f(x) \in F[x]$ е неразл. над $F \Leftrightarrow f(\alpha x+\beta)$ е неразл. над F

$\varphi: F[x] \to F[x]$

$\varphi(f(x)) = f(\alpha x+\beta)$ (коректност – очевидна)

φ е хом. на пръстени?

1) $\varphi(f(x)+g(x)) = \varphi((f+g)(x)) = (f+g)(\alpha x+\beta) = f(\alpha x+\beta) + g(\alpha x+\beta) = \varphi(f(x)) + \varphi(g(x))$

2) $\varphi(f(x) \cdot g(x)) = \varphi((fg)(x)) = (fg)(\alpha x+\beta) = \ldots = \varphi(f(x)) \cdot \varphi(g(x))$ ✓

φ е биекция $\Leftrightarrow \exists \varphi^{-1}: F[x] \to F[x]$ $\varphi^{-1}\varphi = \varphi\varphi^{-1} = id$

$\varphi^{-1}(f(x)) = f(\alpha^{-1}x - \alpha^{-1}\beta)$

$\varphi\varphi^{-1}(f(x)) = \varphi(f(\alpha^{-1}x-\alpha^{-1}\beta)) = f(\alpha\alpha^{-1}x - \alpha^{-1}\beta) + \beta) = f(\alpha\alpha^{-1}x - \alpha\alpha^{-1}\beta + \beta) = f(x) = id(f(x))$

$\varphi^{-1}\varphi(f(x)) = \varphi^{-1}(f(\alpha x+\beta)) = f(\alpha^{-1}(\alpha x+\beta) - \alpha^{-1}\beta) = f(\alpha^{-1}\alpha x + \alpha^{-1}\beta - \alpha^{-1}\beta) = f(x) = id(f(x))$

$\Rightarrow \varphi$ е автоморфизъм (изоморфизъм, действащ в едно и също м-во)

Нека дан, че $f(x)$ е разложим, т.е. $\exists g(x), h(x) \in F[x]: f(x) = g(x)\,h(x)$, $\deg g < \deg f$, $\deg h < \deg f$

$\Rightarrow \varphi(f(x)) = \varphi(g(x)\,h(x)) \overset{\varphi \text{ е хмм}}{=} \varphi(g(x))\,\varphi(h(x))$

$\Rightarrow f(\alpha x+\beta) = g(\alpha x+\beta)\,h(\alpha x+\beta)$ и $\deg f(x) = \deg f(\alpha x+\beta)$, $\deg g(x) = \deg g(\alpha x+\beta)$, $\deg h(x) = \deg h(\alpha x+\beta)$

$\Rightarrow f(\alpha x+\beta)$ е разложим

Нека $f(\alpha x+\beta)$ е разложим, т.е. $\exists t(x), k(x) \in F[x]$

$f(\alpha x+\beta) = t(x)\cdot k(x)$, $\deg t < \deg f(\alpha x+\beta) > \deg h(x)$

$\varphi^{-1}(f(\alpha x+\beta)) = \varphi^{-1}(t(x)\cdot k(x)) = \varphi^{-1}(t(x))\cdot \varphi^{-1}(k(x))$

(φ^{-1} е хмм (аналог. на $\varphi(x)$))

$f(x) = t(\alpha^{-1}x-\alpha^{-1}\beta)\,k(\alpha^{-1}x-\alpha^{-1}\beta)$ и $\deg t(x) = \deg t(\alpha^{-1}x-\alpha^{-1}\beta)$, $\deg k(x) = \deg k(\alpha^{-1}x-\alpha^{-1}\beta)$

$\Rightarrow f(x)$ е разложим

Айзенщайн:

$f(x) = a_0x^n + a_1x^{n-1} + \dots + a_n \in \mathbb{Z}[x]$

Ако $\exists p$ - просто:

1) $p \nmid a_0$

2) $p \mid a_1, \dots, a_n$

3) $p \nmid a_n^2$

$\Rightarrow f(x)$ е неразложим над $\mathbb{Q}$

$f(x)$ е неразл. над $\mathbb{Q} \iff f(\alpha x+\beta)$ е неразл. над $\mathbb{Q}$, $\alpha, \beta \in \mathbb{Z}$

9.8. $f = a_0x^n + \dots + a_n \in \mathbb{Z}[x]$, p-просто, $p \nmid a_0$
и $\bar{f} = \bar{a}_0x^n + \bar{a}_1x^{n-1} + \dots + \bar{a}_n \in \mathbb{Z}_p[x]$

Д-во, че ако $\bar{f}$ е неразл. над $\mathbb{Z}_p$, то f е неразл. над $\mathbb{Q}$

$\mathbb{Z}_p = \mathbb{Z}/(p)$
$\pi: \mathbb{Z} \to \mathbb{Z}_p$
$\pi(a) = \bar{a}$

индуцира $\varphi: \mathbb{Z}[x] \to \mathbb{Z}_p[x]$
$\varphi(f(x)) = \bar{f}(x)$ – φ е хом.
$\deg f = \deg \bar{f}$

? Ако $f \in \mathbb{Z}[x]$ е разл. над $\mathbb{Q} \Rightarrow \bar{f}$ е разл. над $\mathbb{Z}_p$

$\exists g, h \in \mathbb{Z}[x]: f = gh$ и $\deg g < \deg f$, $\deg h < \deg f$

ст.-коеф. на g е b_0 — " — h е c_0 $\Rightarrow p \nmid b_0$ и $p \nmid c_0$
защото $a_0 = b_0 \cdot c_0$

$\varphi(f) = \varphi(gh) = \varphi(g)\varphi(h)$
$\Rightarrow \bar{f} = \bar{g}\bar{h}$ и $\deg \bar{g} = \deg g$, $\deg \bar{h} = \deg h$ $\quad \bar{b} \neq \bar{0}$, $\bar{c} \neq \bar{0}$

9.14. б) $f = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4$, a_i – цели нен. числа, $i = 0, 1, \dots, 4$
Д-во, че f е неразл. над $\mathbb{Q}$

$p = 2$
$\Rightarrow \bar{f} = x^4 + x^3 + x^2 + x + \bar{1} \in \mathbb{Z}_2[x]$

проверка за корени:

$\bar{f}(\bar{0}) = \bar{1}$
$\bar{f}(\bar{1}) = \bar{1}$ ↓

? $f = x^4 + x^3 + x^2 + x + \bar{1} = (x^2 + ax + b)(x^2 + cx + d)$

1 к-и: неопределени коефф.

$x^3 \to$	$a + c = \bar{1}$
$x^2 \to$	$ac + b + d = \bar{1}$ $\Rightarrow ac = \bar{1}$ $\Rightarrow a = c = \bar{1}$
$x^1 \to$	$ad + bc = \bar{1}$ $\Rightarrow a + c = \bar{1}$
$x^0 \to$	$bd = \bar{1}$ $\Rightarrow b = d = \bar{1}$

2 к-и: полиноми от 2-ра ст.

$x^2 \to$ корен $\bar{0}$

$x^2 + x \nearrow$

$x^2 + \bar{1} \to$ корен $\bar{1}$

$x^2 + x + \bar{1} \to$ неразл.

$\Rightarrow$ ако $\bar{f}$ се разлага на произв. на 2 (неразл.) полинома от 2-ра ст., то

$\Rightarrow \bar{f} = x^4 + x^3 + x^2 + x + \bar{1} \not| (x^2 + x + \bar{1})(x^2 + x + \bar{1}) = x^4 + \cancel{2x^3} + \bar{3}x^2 + \cancel{2x} + \bar{1} = x^4 + x^2 + \bar{1}$

б) $f = x^5 + 3x^4 + 2x^3 + x^2 + 4x + 9$

проверка за (целочислени) корени

кандидати: $\pm 1, \pm 3, \pm 9$

$f(1), f(3), f(9) > 0$

	1	3	2	1	4	9
-1	1	2	0	1	3	6
-3	1	0	2	-5	19	
-9	1	-6	56			

$\Rightarrow$ няма корени

$\Rightarrow$ ако f е разложим, то $f = \underbrace{(x^2 + ax + b)}_{g}\underbrace{(x^3 + cx^2 + dx + e)}_{h}$, $a, b, c, d, e \in \mathbb{Z}$ (h неразл.)

6 случая, в зависимост от b и e

редуция по mod 2 или mod 3

$\Rightarrow \bar{f} = \bar{g}\bar{h}$

$\bar{f} = x^5 + x^4 + x^2 + \bar{1} \in \mathbb{Z}_2[x]$

$\bar{f}(\bar{1}) = \bar{0}$

$\bar{f} = x^5 + x^4 + x^2 + \bar{1} \;\big|\; \bar{g} = x + \bar{1}$

$x^5 + x^4 \qquad\qquad \bar{h} = x^4 + x + \bar{1}$ → нередуц.

$x^2 + \bar{1}$

$x^2 + x$

$x + \bar{1}$

$x + \bar{1}$

$\bar{0}$

$f = g\,h$ — II ст. — II ст.

$\bar{f} = \bar{g}\,\bar{h}$ — I ст. — II ст.

$\Rightarrow f$ е нередуц.

$\bar{\bar{f}} = x^5 + \bar{2}x^3 + x^2 + x \in \mathbb{Z}_3[x]$

$\bar{\bar{f}}(\bar{0}) = \bar{0}$

$\Rightarrow \bar{\bar{f}} = x(x^4 + \bar{2}x^2 + x + \bar{1})$

$\bar{\bar{g}}$ $\quad$ $\bar{\bar{h}}$ не зависим от разложение на $\bar{\bar{h}}$

---

2) $f = x^6 + 3x^5 + x^4 + x^3 + 5x^2 + x + 5$

кандидати за корени: $\pm 1, \pm 5$

$f(1), f(5) > 0$

$f(-1) > 0$

$f(-5) > 0$

над $\mathbb{Z}_2$ и $\mathbb{Z}_3$

$\bar{f} = x^6 + x^5 + x^4 + x^3 + x^2 + x + \bar{1}$

$\bar{f}(\bar{0}) = \bar{1}$

$\bar{f}(\bar{1}) = \bar{1}$

? $f = gh$

I ст. II ст.

II ст. I ст.

? $\bar{g} \in \mathbb{Z}_2[x] : \bar{g} \mid \bar{f}$ и $\deg \bar{g} = 2 \Rightarrow \bar{g} = x^2 + x + \bar{1}$

$\bar{f} = x^6 + x^5 + x^4 + x^3 + x^2 + x + \bar{1} \;\big|\; \bar{g} = x^2 + x + \bar{1}$

$x^6 + x^5 + x^4 \qquad\qquad g = x^4 + x$

$x^3 + x^2 + x + \bar{1}$

$x^3 + x^2 + x$

$\bar{1} = \bar{1}$

$\Rightarrow (\bar{f}, \bar{g}) = \bar{1}$

? $\exists h \in \mathbb{Z}_2[x]: \bar{h} \mid \bar{f}$ и $\bar{h}$ е неразл. над $\mathbb{Z}_2$ от $\bar{y}^{-ta}$ ст.

x^3

x^3+x^2

x^3+x

$x^3+\bar{1}$

x^3+x^2+x

$x^3+x^2+\bar{1}$ } корен $\bar{0}$

$x^3+x+\bar{1}$ } неразл.

$x^3+x^2+x+\bar{1} \to$ корен $\bar{1}$

$\bar{f} = x^6+x^5+x^4+x^3+x^2+x+1$

$-\underline{x^6+x^5 \quad\quad x^3}$

$x^4+x^2+x+\bar{1}$

$-\underline{x^4+x^3+x \quad\quad x^3+x^2+\bar{1}}$

$-x^3+x^2+\bar{1}$

$\bar{0}$

$\bar{h} = x^3+x^2+\bar{1}$

$\bar{t} = x^3+x+\bar{1}$

$\Rightarrow \bar{f} = \bar{h} \cdot \bar{t}$

$\mathbb{Z}_3 \to \bar{\bar{f}} = x^6+x^4+x^3+2x^2+x+\bar{2}$

$\bar{\bar{f}}(\bar{0}) = \bar{2}$

$\bar{\bar{f}}(\bar{1}) = \bar{2}$ $\Rightarrow$ няма корени

$\bar{\bar{f}}(\bar{2}) = \bar{1}$

полиноми от $\mathbb{Z}_3[x]$ от

$x^2 \to \bar{0}$

$x^2+x \to \bar{0}$

$x^2+\bar{2}x \to \bar{0}$

$x^2+\bar{1} \to$ неразл.

$x^2+\bar{2} \to \bar{1}$

$x^2+x+\bar{1} \to \bar{1}$

$x^2+2x+\bar{1} \to \bar{2}$

$x^2+x+\bar{2} \to$ неразл.

$x^2+\bar{2}x+\bar{2}$ → неразл.

$\bar{f} = x^6+x^4+x^3+2x^2+x+\bar{2}$ | $\bar{g} = x^2+\bar{1}$

$-x^6+x^4$

$x^3+\bar{2}x^2+x+\bar{2}$

$-x^3 \quad +x$

$\bar{2}x^2+\bar{2}$

$-\bar{2}x^2+\bar{2}$

$\bar{0}$

$\bar{h} = x^4+x+\bar{2}$

$\bar{f} = \bar{g}\cdot\bar{h}$

$\bar{4}\ \ \bar{4}$ ✓

$\mathbb{Z}_2[x]/(x^2+x+\bar{1})$ → F - крайно поле с 8 ел.

↓

$f(x) = ax^2+bx+c$ → 8 ел.

$a^{\wedge}1 \quad b^{\wedge}1 \quad c^{\wedge}1$

9.12. Да се, че f е неразложим над $\mathbb{Q}$

б) $f = x^4-8x^3+21x^2-11x-11$

$f(x+\beta)$? - неразл. над $\mathbb{Q}$?

$f(x+\beta) = (x+\beta)^4-8(x+\beta)^3+21(x+\beta)^2-11(x+\beta)-11$

$= x^4+4\beta x^3+6\beta^2x^2+4\beta^3x+\beta^4-8x^3-24\beta x^2-24\beta^2x-8\beta^3+21x^2+42\beta x+21\beta^2-11x-11\beta-11$

$= x^4+4(\beta-2)x^3+(6\beta^2-24\beta+21)x^2+(4\beta^3-24\beta^2+42\beta-11)x+\beta^4-8\beta^3+21\beta^2-11\beta-11$

$\beta \overset{?}{=} 2$

$\Rightarrow f(x+2) = x^4-3x^2+9x+3$

за $p=3$ Айзенщайн $f(x+2)$ е неразл. над $\mathbb{Q} \Rightarrow f(x)$ е неразл. над $\mathbb{Q}$

г) $f(x) = x^p-px+2p-1$, p - просто

$f(x+\beta) = (x+\beta)^p-p(x+\beta)+2p-1 = x^p+\binom{p}{1}x^{p-1}\beta+\dots+\binom{p}{p-1}x\beta^{p-1}+\beta^p-px-p\beta+2p-1$

$\binom{p}{1} = p$

$f(x+p)$ е керазл. над $\mathbb{Q}$, ако $p \mid p^p - pp + 2p - 1$

$4 \mid p^2 \mid p^p - pp + 2p - 1$

$p^p - pp + 2p - 1 \equiv 0 \pmod p$

$\Leftrightarrow p^p \equiv 1 \pmod 5$

$\Rightarrow p = \pm 1$

1) $p = -1 < 0 \Leftrightarrow p = 2$

$f(x) = x^2 - 2x + 3$ керазл.

2) $p = 1 \Rightarrow$ св. коеф.: $1^p - p + 2p - 1 = p$

$p \mid p^2$

$\Rightarrow$ керазл. над $\mathbb{Q}$

21.05.

Първите зад. - на снимки!

6.4. $f = x^4 + x^3 - 2x^2 + \mu x + 21 \in \mathbb{C}[x]$, $\mu = ?$

$x_1 + x_2 = x_3 x_4 = t$

Виет:

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = -1 \\ x_1x_2 + x_1x_3 + x_1x_4 + x_2x_3 + x_2x_4 + x_3x_4 = -2 \\ x_1x_2x_3 + x_1x_2x_4 + x_1x_3x_4 + x_2x_3x_4 = -\mu \\ x_1 \cdot x_2 \cdot x_3 \cdot x_4 = 21 \end{cases}$$

2) $\Rightarrow x_1x_2 + \underbrace{(x_1 + x_2)}_{t}(x_3 + x_4) + \underbrace{x_3x_4}_{t} = -2$

$x_1x_2 + (x_3 + x_4)t + t = -2$

1) $\Rightarrow x_3 + x_4 = -1 - t$

4) $\Rightarrow x_1x_2 \cdot t = 21$

$\Rightarrow x_1 \cdot x_2 = -2 - t + (1+t) \cdot t = t^2 - 2$

$\Rightarrow t^3 - 2t - 21 = 0$

3) $\Rightarrow \underbrace{x_1 x_2}_{t^2-2}(\underbrace{x_3 + x_4}_{-1-t}) + \underbrace{(x_1 + x_2)}_{t} \underbrace{x_3 x_4}_{t} = -\mu$

$-t^3 - t^2 + 2t + 2 + t^2 = -\mu \;/\cdot(-1)$

$t^3 - 2t - 2 = \mu \Rightarrow \boxed{\mu = 19}$

$\Leftarrow \mu = 19 \overset{?}{\Rightarrow} x_1 + x_2 = x_3 x_4 = t$

$f = x^4 + x^3 - 2x^2 + 19x + 21$

$f = (x^2 - \underbrace{(x_1 + x_2)}_{t} x + x_1 x_2)(x^2 - (x_3 + x_4)x + \underbrace{x_3 x_4}_{t})$

$f \overset{?}{=} (x^2 - 3x + 7)(x^2 + 4x + 3)$

$f = x^4 + x^3 - 2x^2 + 19x + 21 \quad \big| \; x^2 - 3x + 7$

$-\,x^4 - 3x^3 + 7x^2 \qquad\qquad x^2 + 4x + 3$

$4x^3 - 9x^2 + 19x + 21$

$-\,4x^3 - 12x^2 + 28x$

$3x^2 - 9x + 21$

$-\,3x^2 - 9x + 21$

0

$f = a_0 x^n + \dots + a_n \in F[x], \; F \leq L$

$\alpha \in L$ е k-кратен корен, ако $f(x) = a_0 (x-\alpha)^k \cdot g(x), \; g(\alpha) \neq 0 \quad (k \in \mathbb{N}, k \geq 1)$

$\operatorname{char} F = 0$ Дадено α да е k-кр. корен.

$f(\alpha) = f'(\alpha) = \dots = f^{(k-1)}(\alpha) = 0, \; f^{(k)}(\alpha) \neq 0$

- $\mathrm{char}\, F = 0$ α е к-кр. корен $\Leftrightarrow f(\alpha) = f'(\alpha) = 0$

6.8. Да се опр. кратността на корен α за пол. f ($\mathrm{char}\, F = 0$)
$\alpha = 1$, $f = x^{2n+1} - (2n+1)x^{n+1} + (2n+1)x^n - 1$, $n \in \mathbb{N}$

$f(1) = 1 - 2n - 1 + 2n + 1 - 1 = 0$
$f'(x) = (2n+1)x^{2n} - (2n+1)\cdot(n+1)\cdot x^n + n(2n+1)x^{n-1}$
$f'(1) = (2n+1)\cdot(1 - (n+1) + n) = 0$
$f''(1) = (2n+1)\cdot n\,(2 - (n+1) + (n-1)) = 0$
$f'''(x) = (2n+1)\,(n(2(2n-1)))\,x^{2n-2} - (n+1)(n-1)x^{n-2} + (n-1)(n-2)x^{n-3})$
$f'''(1) = (2n+1)\,n\,(4n - 2n^2 + 1 + n^2 - 3n + 2) > 0$

$\Rightarrow \alpha = 1$ е тричкратен

6.9. ДСК $a, b \in F$, за които 1 е двукратен корен на пол.
$f = ax^{n+1} + bx^n + 1$, $\mathrm{char}\, F = 0$, $n \in \mathbb{N}$

$$\begin{cases} f(1) = a + b + 1 = 0 \\ f'(1) = (n+1)a + nb = 0 \\ f''(1) = (n+1)\cdot n \cdot a + n(n-1)\cdot b \neq 0 \end{cases}$$

$$\xrightarrow{-n} \begin{pmatrix} 1 & 1 & | & -1 \\ n+1 & n & | & 0 \end{pmatrix} \xrightarrow{-1} \begin{pmatrix} 1 & 1 & | & -1 \\ 1 & a & | & n \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & | & -n-1 \\ 1 & a & | & n \end{pmatrix}$$

$\Rightarrow a = n$, $b = -n-1$

$f''(1) = (n+1)n^2 - n(n-1)(n+1) = n^3 + n^2 - n^3 + n = n^2 + n > 0$
$\qquad\qquad\qquad\qquad\quad \underbrace{\qquad\qquad}_{n^2-1}$

6.10. $\lambda = ?$; $f = x^4 + \lambda x^3 + 27 \in \mathbb{C}[x]$ има кратен к

Нека α е кр. корен на f

$f(\alpha) = \alpha^4 + \lambda\alpha^3 + 27 = 0$

$f'(\alpha) = 4.\alpha^3 + 3\lambda.\alpha^2 = 0$

$\alpha^2(4\alpha + 3\lambda) = 0$

$\alpha = 0$ $\Rightarrow$ $\alpha = -\frac{3}{4}\lambda$

$f(0) = 27 \neq 0$

$f(-\frac{3}{4}\lambda) = (-\frac{3}{4}\lambda)^4 + \lambda(-\frac{3}{4}\lambda)^3 + 27 = 0 = \frac{81}{256}\lambda^4 - \frac{27}{64}\lambda^4 + 27$ $/:27$

$\lambda^4 = 256$

$\lambda_{1,2} = \pm 4$, $\lambda_{3,4} = \pm 4i$

6.11. ДСК ст-стите на λ и μ, за които

$f = 3x^5 - 5\lambda x^4 + 5\mu x - 3 \in \mathbb{C}[x]$ има прост. корен

Нека α е прост. корен

$\Rightarrow f(\alpha) = 3\alpha^5 - 5\lambda\alpha^4 + 5\mu\alpha - 3 = 0$

$f'(\alpha) = 15\alpha^4 - 20\lambda\alpha^3 + 5\mu = 0$

$f''(\alpha) = 60\alpha^3 - 60\lambda\alpha^2 = 0$

$f^{(4)}(\alpha) = 180\alpha^2 - 120\lambda\alpha = 0$

$:60$

$\alpha^2(\alpha - \lambda) = 0$

$\alpha = 0 \Rightarrow \alpha = \lambda$

$f(0) = -3 \neq 0$

$\Rightarrow f'(\lambda) = 15\lambda^4 - 20\lambda^4 + 5\mu = 0 \Leftrightarrow \boxed{\mu = \lambda^4}$

$f(\lambda) = 3\lambda^5 - 5\lambda^5 + 5\lambda^5 - 3 = 0 \Rightarrow \boxed{\lambda^5 = 1} \Rightarrow \lambda_k = \cos\frac{2k\pi}{5} + i\sin\frac{2k\pi}{5}$

$k = 0, 1, 2, 3, 4$

$\mu = \lambda^4 = \lambda^{-1} \Rightarrow \mu_k = \lambda_{5-k}, \quad \lambda_5 = \lambda_0 = 1$

$\lambda_k = \lambda_1$

$k = 0, 1, 2, 3, 4$

6.12. Нека $\mathrm{char}\, F = 0$. Да се докаже, че

$f = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}$ няма кр. корени

Доказателство: Нека α е кр. корен

$f(\alpha) = 1 + \frac{\alpha}{1!} + \frac{\alpha^2}{2!} + \dots + \frac{\alpha^n}{n!} = 0$

$f'(\alpha) = 1 + \alpha + \dots + \frac{\alpha^{n-1}}{(n-1)!} = 0$

$\Rightarrow f(\alpha) - f'(\alpha) = 0 \Leftrightarrow \alpha = 0 \Leftrightarrow f(0) = 0$, но $f(0) = 1 \neq 0$

$\Rightarrow$ няма кр. корени

6.14. ОСН ост. при дел. на $f \in F[x]$ с нат. $g = (x-\alpha)(x-\beta)$, $\alpha, \beta \in F$

$f = g \cdot q + r$, $\deg r < \deg g = 2$

$\Rightarrow r = ax + b$, $a, b \in F$

$f(\alpha) = g(\alpha) q(\alpha) + r(\alpha) = a\alpha + b$

$g(\alpha) = 0$

1 сл. Пусть $\alpha \neq \beta$

$f(\beta) = g(\beta) q(\beta) + r(\beta) = a\beta + b$

$g(\beta) = 0$

$f(\alpha) - f(\beta) = a(\alpha - \beta)$

$\Rightarrow a = \dfrac{f(\alpha) - f(\beta)}{\alpha - \beta}$ $\Rightarrow b = f(\alpha) - a\alpha = \dfrac{\alpha f(\alpha) - \beta f(\alpha) - \alpha f(\alpha) + \alpha f(\beta)}{\alpha - \beta}$

$= \dfrac{\alpha f(\beta) - \beta f(\alpha)}{\alpha - \beta}$

$\Rightarrow r = ax + b = \ldots$

2 сл. $\alpha = \beta$ $\quad g = (x-\alpha)^2$

$f'(\alpha) = g'(\alpha) \cdot q(\alpha) + g(\alpha) \cdot q'(\alpha) + r'(\alpha) = r'(\alpha) = a$

$g'(\alpha) = 0$, $g(\alpha) = 0$

$\Rightarrow b = f(\alpha) - a\alpha = f(\alpha) - \alpha f'(\alpha)$

$\Rightarrow r = f'(\alpha)x + f(\alpha) - \alpha f'(\alpha)$

6.14. ДСИ остатък при дел-на пол. f с пол. g, когато

а) $f = x^n - 1$, $g = x^3 - 2x^2 + x$

б) $f = x^n - 2x^{n-1} + 2x$, $g = x^3 - 5x^2 + 8x - 4$

а) $f = gq + r$, $\deg r < \deg g = 3$

$r = ax^2 + bx + c$

$g = x(x^2 - 2x + 1) = x(x-1)^2$

$f(0) = g(0)\,q(0) + r(0) = c \Rightarrow \boxed{c = -1}$

$-1 =$ 0

$0 = f(1) = g(1)\cdot q(1) + r(1) = a + b + c \Rightarrow \boxed{a + b = 1}$

$c = -1$

$\Rightarrow a = n - 1$, $b = 2 - n$

$n = f'(1) = g'(1)\,q(1) + g(1)\,q'(1) + r'(1) = 2a + b \Rightarrow \boxed{2a + b = n}$

$g(1) = 0$, $g'(1) = 0$

$\Rightarrow r = (n-1)x^2 + (2-n)x - 1$

28.05. Помощна зад.

За кои цели неотрицателни числа m, n и p е дел. f, където:

а) $g = x^2 + x + 1$, $f = x^{3m+2} + x^{3n+1} + x^{3p}$

б) $g = x^2 - x + 1$, $f = x^{3m+2} + x^{3n+1} + x^{3p}$ $(\mathrm{char}\, F \neq 2)$

$f \equiv h \pmod g$, ако $g \mid (f - h)$

$\equiv$ – рел. на еквив.

+ още св-ва

$f \equiv 0 \pmod g$

а)

$x^3-1=(x-1).g \Rightarrow x^3 \equiv 1 \pmod g$

$f = \underset{1(g)}{x^{3m}} x^2 + \underset{1(g)}{x^{3n}} x + \underset{1(g)}{x^{3p}} \equiv x^2+x+1 \pmod g \equiv 0 \pmod g$

$\forall m,n,p \in \mathbb{N} \cup \{0\} : g \mid f$

б) $x^3+1=(x+1).g \equiv 0 \pmod g \rightsquigarrow x^3 \equiv -1 (g)$

$f = \underset{(-1)^m(g)}{x^{3m}} x^2 + \underset{(-1)^n(g)}{x^{3n}} x + \underset{(-1)^p(g)}{x^{3p}} \equiv (-1)^m x^2 + (-1)^n x + (-1)^p \pmod g$

$\Rightarrow g \mid f \Leftrightarrow m, n$ и p са с еднаква четност

6.15. $\operatorname{char} F = 0$

Дск $\forall n \in \mathbb{N}: f=(x+1)^n + x^n + 1$ се дели на $g=(x^2+x+1)^2$

$h = x^2+x+1 \Rightarrow h^2 = g$

1) $h \mid f$ за кои $n \in \mathbb{N}$?

$x^3 \equiv 1 \pmod h$

$x^2+x+1 \equiv 0 \pmod h$

$x+1 \equiv -x^2 \pmod h$

$\Rightarrow f=(x+1)^n + x^n + 1 \equiv (-1)^n . x^{2n} + x^n + 1 \pmod h \equiv (-1)^{3q+r} x^{6q+3r} + x^{3q+r} + 1 \pmod h$

$n = 3q+r, \quad 0 \le r \le 2$

$f \equiv (-1)^{3q+r} x^{2r} + x^r + 1 \pmod h$

I сл. $r=0$

$f \equiv (-1)^{3q} + 1 + 1 \pmod h \not\equiv 0 \pmod h, \forall q \in \mathbb{N}$ ↯

Ч ca. $5=1$

$f \equiv (-1)^{3q+1} \cdot x^2+x+1 \pmod h$

Ч. 1. $q=2k \Rightarrow f \equiv \underbrace{(-1)^1 x^2+x+1}_{x+1 \,(h)} \pmod h \equiv 2x+2 \pmod h \not\equiv 0 \pmod h$ ↯

Ч. 2. $q=2k+1 \Rightarrow f \equiv x^2+x+1 \pmod h \equiv 0 \pmod h$ ✓

Ч ca. $5=2$

$f \equiv (-1)^{3q+2} \underbrace{x^4}_{x \,(h)} + x^2+1 \pmod h \equiv x^2 + (-1)^{3q+2} x+1 \pmod h$

Ч. 1. $q=2k$: $f \equiv x^2+(-1)^2 x+1 \pmod h \equiv 0 \pmod h$ ✓

Ч. 2. $q=2k+1$: $f \equiv x^2+(-1)^5 x+1 \,(h) \not\equiv 0 \,(h)$ ↯

$\Rightarrow h \mid f \Leftrightarrow n=3(2k+1)+1 = 6k+4$
$\Leftrightarrow n=3(2k)+2 = 6k+2$

2) $h^2 \mid f \Leftrightarrow f(x)=(x^2+x+1)^2 \cdot k(x)$

$f' = n(x+1)^{n-1} + nx^{n-1} = 2(x^2+x+1)\cdot(2x+1)\cdot k(x) + (x^2+x+1)^2 \cdot k'(x) \Leftrightarrow \underbrace{h}_{x^2+x+1} \mid f'$

$\underbrace{(-1)^{n-1}}_{-1} \cdot x^{2(n+1)} \,(h)$

? $n \in \mathbb{N}$: $f' \equiv 0 \pmod h$

I ca. $n=6k+2$

$f' \equiv (6k+2)\cdot(-1)^{6k+2-1} \cdot x^{12k+2} + (6k+2) \cdot x^{6k+2-1} \pmod h \;/:(6k+2)$

$f' \equiv \underbrace{-x^2+x}_{x+1 \,(h)} \pmod h \equiv 2x+1 \pmod h \not\equiv 0 \pmod h$ ↯

Ч ca. $n=6k+4$

$f' \equiv (6k+4)\cdot(-1)^{6k+3} \cdot x^{2\cdot(6k+3)} + (6k+4)\, x^{6k+4-1} \pmod h \;/:(6k+4)$

$\underbrace{1 \pmod h}_{1 \,(h)}$

$f' \equiv -1+1 \pmod h \equiv 0 \pmod h$ ✓

$f(x_1, x_2, \dots, x_n) = F[x_1, x_2, \dots, x_n]$

$a x_1^{k_1} x_2^{k_2} \cdots x_n^{k_n}$, $a \in F$

↳ едночлен

$\deg f(x_1, \dots, x_n)$ = степен на едночлен от най-висока степен

лексикографска наредба

$u = a x_1^{k_1} x_2^{k_2} \cdots x_n^{k_n}$

$v = b x_1^{l_1} x_2^{l_2} \cdots x_n^{l_n}$ $\quad k_i, l_i \geq 0$, $a, b \in F$

$u > v$, ако за някое $i \in \{1, \dots, n\}$ е в сила: $k_1 = l_1, \dots, k_{i-1} = l_{i-1}$ и $k_i > l_i$

$x_1^3 x_2 > x_1 x_2^5 > x_1 x_3^6 \dots$

Лема за главен едночлен: (старши) ако $f(x_1, \dots, x_n) \in F[x_1, \dots, x_n]$ е със старши едночл. $u = a x_1^{k_1} \cdots x_n^{k_n}$, и $g(x_1, \dots, x_n)$ — " — $v = b x_1^{l_1} \cdots x_n^{l_n}$, то ст. едночл. на $f.g$ е $uv = ab x_1^{k_1+l_1} \cdots x_n^{k_n+l_n}$

Симетр. полиноми

$\forall \tau \in S_n$

$f(x_1, \dots, x_n)$ е сим. пол $\Leftrightarrow f(x_1, \dots, x_n) = f(x_{\tau(1)}, \dots, x_{\tau(n)})$

Елементарни сим. пол:

$\sigma_1 = x_1 + x_2 + \cdots + x_n$

$\sigma_2 = x_1 x_2 + x_1 x_3 + \cdots + x_{n-1} x_n = \sum_{i<j} x_i x_j$

$\vdots$

$\sigma_n = x_1 \cdot x_2 \cdot \cdots \cdot x_n$

Осн. т-ма на сим. пол:

$F[x_1, \dots, x_n]$ - обл. на цялост, $f(x_1, \dots, x_n) \in F[\dots]$ - сим.

Тогава $\exists g \in F[\dots]$, т.ке

$f(x_1, \dots, x_n) = g(\sigma_1, \sigma_2, \dots, \sigma_n)$

7.1. а) Да се изрази чрез елементарни сим. пол. $\sigma_1, \sigma_2, \sigma_3$ симетр.-ф-ция Σ:

$\Sigma = x_1^4 x_2^2 + x_1^4 x_3^2 + x_1^2 x_2^4 + x_1^2 x_3^4 + x_2^4 x_3^2 + x_2^2 x_3^4$

Бел: $a x_1^{l_1} \cdots x_n^{l_n}$ ↓ $b x_1^{l_1} \cdots x_n^{l_n} : l_1 + l_2 + \cdots + l_n = 2l_1 + \cdots + l_n$, $l_1 \geq l_2 \geq \cdots \geq l_n$

$4\,2\,0 \to 1 \cdot \sigma_1^{4-2}\sigma_2^{2-0}\sigma_3^0 = \sigma_1^2\sigma_2^2$

$f_1 = \sigma_1^2\sigma_2^2 = (x_1+x_2+x_3)^2(x_1x_2+x_1x_3+x_2x_3)^2 = x_1^4x_2^2 + \cdots$

$4\,1\,1 \to A \cdot \sigma_1^{4-1}\sigma_2^{1-1}\sigma_3^{1-0} = A\cdot\sigma_1^3\sigma_3$

$3\,3\,0 \to B \cdot \sigma_1^{3-3}\sigma_2^{3-0}\sigma_3^0 = B\sigma_2^3$

$3\,2\,1 \to C\sigma_1^{3-2}\sigma_2^{2-1}\sigma_3^{1-0} = C\sigma_1\sigma_2\sigma_3$

$2\,2\,2 \to D\sigma_1^{2-2}\sigma_2^{2-2}\sigma_3^{2-0} = D\sigma_3^2$

$\Sigma = \sigma_1^2\sigma_2^2 + A\sigma_1^3\sigma_3 + B\sigma_2^3 + C\sigma_1\sigma_2\sigma_3 + D\sigma_3^2$

$\sigma_1 = x_1+x_2+x_3$

$\sigma_2 = x_1x_2+x_1x_3+x_2x_3$

$\sigma_3 = \sigma_1\sigma_2\sigma_3$

1) $x_1 = x_2 = 1$, $x_3 = 0$

$\sigma_1 = 2$, $\sigma_2 = 1$, $\sigma_3 = 0$

$2 = \Sigma = 4 + B \Rightarrow \boxed{B = -2}$

2) $x_1 = x_2 = 1$, $x_3 = -2$

$\sigma_1 = 2$, $\sigma_2 = -3$, $\sigma_3 = -2$

$2 + 32 + 8 = \Sigma = (-2)\cdot(-3)^3 \cdot 4D = 54 + 4D \Rightarrow \boxed{D = -\frac{12}{4} = -3}$

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3) $x_1 = 1$, $x_2 = x_3 = -2$

$\sigma_1 = -3$, $\sigma_2 = 0$, $\sigma_3 = 4$

$\underbrace{128+32+8}_{168} = \Sigma = -27.4.A - 3.16 = -108A - 48 \Rightarrow \boxed{A=-2}$

4) $x_1 = x_2 = x_3 = 1$

$r_1 = 3, r_2 = 3, r_3 = 1$

$6 = \Sigma = 81 - 54 - 54 + 9C - 3 = -30 + 9C \Rightarrow \boxed{C=4}$

$\Sigma = (x_1+x_2-x_3-x_4)(x_1-x_2+x_3-x_4)(x_1-x_2-x_3+x_4)$

$x_1^3 - 300 - r_1^3$

$210 - Ar_1r_2$

$111 - Br_3$

Ф.5. Да се изрази чрез p и q (коефициентите им съответно) симетричната функция Σ от корените x_1, x_2, x_3 на $f = x^3 + px + q$

а) $\Sigma = (x_1^2+x_1x_2+x_2^2)(x_1^2+x_1x_3+x_3^2)(x_2^2+x_2x_3+x_3^2)$

$\underbrace{(x_1+x_2)}_{-x_3}{}^2 - x_1x_2 = x_3^2 - x_1x_2 = \cancel{x_3^2} - p + (-x_3)x_3 = -p$

$p - \underbrace{(x_1+x_2)}_{-x_3} \cdot x_3$

Виет:

$x_1+x_2+x_3 = r_1 = 0$

$x_1x_2+x_1x_3+x_2x_3 = r_2 = p$

$x_1x_2x_3 = r_3 = -q$

$\Rightarrow \Sigma = -p^3$

б) $\Sigma = \frac{\overbrace{1+x_1}}{x_1} + \frac{\overbrace{1+x_2}}{x_2} + \frac{\overbrace{1+x_3}}{x_3}$

$= \frac{\overbrace{x_1+x_2+x_3}^{=a} + x_1^2+x_2^2+x_3^2}{(1+x_1)(1+x_2)(1+x_3)} = \frac{\Sigma_1}{\Sigma_2}$

$\Sigma_2 = 1 + \underbrace{(x_1+x_2+x_3)}_{a} + \underbrace{(x_1x_2+x_1x_3+x_2x_3)}_{p} + \underbrace{x_1x_2x_3}_{-q} = 1+p-q$

$\Sigma_1 = x_1^2+x_2^2+x_3^2 = \underbrace{(x_1+x_2+x_3)}_{a}^2 - 2\underbrace{(x_1x_2+x_1x_3+x_2x_3)}_{b} = -2p$

$\Rightarrow \Sigma = \frac{2p}{q-p-1}$

б) $\Sigma = \frac{x_1^2}{1+x_1} + \frac{x_2^2}{1+x_2} + \frac{x_3^2}{1+x_3}$ $\qquad \frac{x_i^2}{1+x_i} = \frac{x_i^2-1}{1+x_i} + \frac{1}{1+x_i} = x_i - 1 + \frac{1}{1+x_i}$

$\Sigma = \underbrace{x_1+x_2+x_3}_{0} - 3 + \sum_{i=1}^{3} \frac{1}{1+x_i}$

Помощна зад. (7.3.): Нека f е пол. със ст. коеф. 1 и корени $x_1, \dots, x_n$, $a \in F$, $f(a) \neq 0$

Да се докаже:

а) $\prod_{i=1}^{n} (a-x_i) = f(a)$

г) $\sum_{i=1}^{n} \frac{1}{a-x_i} = \frac{f'(a)}{f(a)}$

в) $\sum_{i=1}^{n} \frac{1}{(a-x_i)^2} = \frac{(f'(a))^2 - f(a).f''(a)}{(f(a))^2}$

$f(x) = (x-x_1).(x-x_2).\dots.(x-x_n) = \prod_{i=1}^{n}(x-x_i)$

а) $\Rightarrow f(a) = \prod_{i=1}^{n} a-x_i$

г) $f'(x) = (x-x_2)\dots(x-x_n) + (x-x_1)(x-x_3)\dots(x-x_n) + \dots + (x-x_1)\dots(x-x_{n-1})$

$\Rightarrow \frac{f'(a)}{f(a)} = \sum_{i=1}^{n} \frac{1}{a-x_i}$

в) $\left(\frac{f'(a)}{f(a)}\right)' = \frac{f''(a).f(a) - (f'(a))^2}{(f(a))^2}$ ✓

$\left(\sum_{i=1}^{n} \frac{1}{a-x_i}\right)' = -\sum_{i=1}^{n} \frac{1}{(a-x_i)^2}$

Продължаме нас дз:

$f(-1) = -1-p+q$

$f'(-1) = 3+p$

$$\Sigma = -3 + \sum_{i=1}^{3} \frac{1}{1+x_i} = -3 - \frac{f'(-1)}{f(-1)} = -3 + \frac{3+p}{1+p-q}$$

2) $$\Sigma = \frac{x_1}{(1-x_1)^2} + \frac{x_2}{(1-x_2)^2} + \frac{x_3}{(1-x_3)^2}$$

$$\frac{x_i}{(1-x_i)^2} = \frac{x_i - 1}{(1-x_i)^2} + \frac{1}{(1-x_i)^2} = -\frac{1}{1-x_i} + \frac{1}{(1-x_i)^2}$$

$$\Sigma = -\sum_{i=1}^{3} \frac{1}{1-x_i} + \sum_{i=1}^{3} \frac{1}{(1-x_i)^2} = -\frac{f'(1)}{f(1)} + \frac{(f'(1))^2 - f(1)f''(1)}{(f(1))^2}$$

$$= -\frac{3+p}{1+p+q} + \frac{(3+p)^2 - 6(1+p+q)}{(1+p+q)^2}$$

g) $$\Sigma = \frac{x_1^2}{(x_2-x_3)^2} + \frac{x_2^2}{(x_1-x_3)^2} + \frac{x_3^2}{(x_1-x_2)^2}$$

$$\frac{x_1^2}{(x_2-x_3)^2} = \frac{x_1^2}{(x_2+x_3)^2 - 4x_2x_3} = \frac{x_1^2}{x_1^2 - 4x_2x_3} = \frac{x_1^2}{x_1^2 + 4q} = \frac{x_1^3}{x_1^3 + 4qx_1}$$

($-x_1$; $-4x_2x_3 = -\frac{4q}{x_1} \neq 0$)

$f(x_1) = x_1^3 + px_1 + q = 0$

$x_1^3 = -px_1 - q$

$$= \frac{-px_1 - q + 4q - 4q}{-px_1 + 3q} = 1 - \frac{4q}{-px_1 + 3q} = 1 - \frac{4q}{p} \cdot \frac{1}{\frac{3q}{p} - x_1}$$

$x_1 = 0$

$x_2 = -x_3$

$\Rightarrow \Sigma = 2$

$$\Rightarrow \Sigma = 3 - \frac{4q}{p} \cdot \sum_{i=1}^{3} \frac{1}{\frac{3q}{p} - x_i} = 3 - \frac{4q}{p} \cdot \frac{f'\left(\frac{3q}{p}\right)}{f\left(\frac{3q}{p}\right)} = -\frac{\quad}{\quad}$$

04.06.

5.7. p - просто. ДСК, че а) броят на неразл. над $\mathbb{Z}_p$ полиноми от вида $f = x^2 + ax + b$ е равен на $\frac{p(p-1)}{2}$

б) $g = x^3 + ax^2 + bx + c$ - брой неразл.

общ. бр. полиноми: p^2
от x^{p-p} ст., ст. коеф $=1$

разложими: $f=(x-\alpha)(x-\beta)$, $\alpha\neq\beta\in\mathbb{Z}_p$ – брой $=\binom{p}{2}=\frac{p(p-1)}{2}$

$f=(x-\alpha)^2$ – брой $=p$

брой неразл. $=p^2-\left(\frac{p(p-1)}{2}+p\right)=\frac{2p^2-p^2+p-2p}{2}=\frac{p(p-1)}{2}$ ✓

б) $g=x^3+ax^2+bx+c$

общ. бр. пол $=p^3$

разложими: $g=(x-\alpha)(x-\beta)(x-\gamma)$, $\alpha\neq\beta\neq\gamma\in\mathbb{Z}_p$ – брой $=\binom{p}{3}=\frac{p\cdot(p-1)\cdot(p-2)}{6}$

$g=(x-\alpha)^2(x-\beta)$, $\alpha\neq\beta\in\mathbb{Z}_p$ – брой $=p(p-1)$

$g=(x-\alpha)^3$, брой $=p$

$g=(x-\alpha)\cdot(x^2+ax+b)$, брой $=p\cdot\left(p-\frac{(p-1)}{2}\right)$
↓ неразл.

⇒ бр. неразл. $=p^3-$ сбора на разл.

R – комут. пр. с 1, $I\lhd R$.
Дсд, че $I[x]\leq R[x]$ и $R[x]/I[x]\cong(R/I)[x]$

$\bar{u}:R\to R/I$ – ест. хом (сюрективен)
$\bar{u}(a)=a+I=\bar{a}$, $\forall a\in R$
$\ker\bar{u}=\{a\in R\mid\bar{u}(a)=\bar{0}=I\}=I$

и индуцира хом: $\varphi: R[x] \to (R/I)[x]$

$\varphi(f(x)) = \overline{f}(x)$ $\quad$ φ е сюрекция

$f(x) = a_0x^n + a_1x^{n-1} + \dots + a_n$

$\overline{f}(x) = \bar{a}_0x^n + \bar{a}_1x^{n-1} + \dots + a_n$

$\bar{a}_i = a_i + I$

$\operatorname{Ker}\varphi = \{ f \in R[x] \mid \varphi(f) = \bar{0} \} = I[x]$

$\Leftrightarrow a_i \in I, \ \forall i \in \{0, \dots, n\}$

$\Rightarrow$

$I[x] \trianglelefteq R[x]$ и $R[x]/I[x] \cong (R/I)[x]$

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p - ест. число: $\mathbb{Z}[x]/(p) \cong \mathbb{Z}_p[x]$ (област, но не е поле)

ако p-просто $\Rightarrow (p)$ - прост идеал, но не е max

5.13. а) $\mathbb{Q}[x]/(x^2-2) \cong \mathbb{Q}(\sqrt{2})$

б) $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$

$f, g \in \mathbb{Q}[x]$, $I = (x^2-2)$ $\Rightarrow f + I = g + I \Leftrightarrow f - g \in I \Leftrightarrow f \equiv g \pmod{(x^2-2)}$

$\Leftrightarrow f$ и g дават еднакви остатъци при деление с x^2-2

$\Rightarrow f = (x^2-2).q + r$, $\deg r < \deg(x^2-2)$

$\Rightarrow r = ax + b$ - елем. на $\mathbb{Q}[x]/I$

x^2-2 е неразложим над $\mathbb{Q} \Rightarrow \mathbb{Q}[x]/I$ е поле

$\varphi : \mathbb{Q}[x] \to \mathbb{Q}(\sqrt{2})$

$\varphi(f(x)) = f(\sqrt{2})$

$f(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_n$

$f(\sqrt{2}) = a_0(\sqrt{2})^n + a_1(\sqrt{2})^{n-1} + \dots + a_n = a + b\sqrt{2} \in \mathbb{Q}(\sqrt{2})$

φ е ХЛМ:

- $\varphi(f(x)+g(x)) = (f+g)(\sqrt{2}) = f(\sqrt{2}) + g(\sqrt{2}) = \varphi(f(x)) + \varphi(g(x))$ ✓

- $\varphi(f(x)\cdot g(x)) = (f\cdot g)(\sqrt{2}) = f(\sqrt{2})\cdot g(\sqrt{2}) = \varphi(f(x))\cdot\varphi(g(x))$ ✓

- φ е сюрекция

$\mathrm{Im}\,\varphi \subseteq \mathbb{Q}(\sqrt{2})$

$b + a\sqrt{2} \in \mathbb{Q}(\sqrt{2}) \Rightarrow \exists f = ax + b : \varphi(f) = b + a\sqrt{2} \Rightarrow \mathbb{Q}(\sqrt{2}) \subseteq \mathrm{Im}\,\varphi$

- $\mathrm{Ker}\,\varphi = \{ f(x) \in \mathbb{Q}[x] \mid \varphi(f) = \bar{0} \} = I = (x^2 - 2)$

$\Leftrightarrow f(\sqrt{2}) = 0$

$\Leftrightarrow f(-\sqrt{2}) = 0$

$f(\sqrt{2}) = a + b\sqrt{2}$

$\Leftrightarrow a = b = 0$

$\Leftrightarrow (x^2-2) \mid f \overset{\mathbb{Q}[x]}{\Leftrightarrow} f \in \mathrm{Ker}\,\varphi \Leftrightarrow (x^2-2) = \mathrm{Ker}\,\varphi$ ($I = (x^2-2)$)

Т-ма за ХММ $\Rightarrow \mathbb{Q}[x]/(x^2-2) \cong \mathbb{Q}(\sqrt{2})$

Зад: Да се докаже, че $\mathbb{Z}_2[x]/(x^2+x+\bar{1})$ - поле с 4 ел. и да се напишат табл. за $+$ и $\cdot$ в него

$x^2 + x + \bar{1}$ - неразл. над $\mathbb{Z}_2[x]$

$\Rightarrow$ поле с ел. $f = \underset{\wedge}{a}x + \underset{\wedge}{b}$, $a, b \in \mathbb{Z}_2$ – 4 ел.

$x^2 + x + \bar{1} = \bar{0}$

$x^2 = x + \bar{1}$

$x = x^2 + \bar{1}$

$\bar{1} = x^2 + x$

$+$	$\bar{0}$	$\bar{1}$	x	$x+\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	x	$x+\bar{1}$
$\bar{1}$	$\bar{1}$	$\bar{0}$	$x+\bar{1}$	x
x	x	$x+\bar{1}$	$\bar{0}$	$\bar{1}$
$x+\bar{1}$	$x+\bar{1}$	x	$\bar{1}$	$\bar{0}$

$\cdot$	$\bar{0}$	$\bar{1}$	x	$x+\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$
$\bar{1}$	$\bar{0}$	$\bar{1}$	x	$x+\bar{1}$
x	$\bar{0}$	x	$x+\bar{1}$	$\bar{1}$
$x+\bar{1}$	$\bar{0}$	$x+\bar{1}$	$\bar{1}$	x

$F =$

$\mathbb{Z}_3[x]/(x^3 - x + \bar{1})$ – поле с 27 ел. ?

$\overline{AB^{-1}} \in F = ?$

$A = x^4 - \bar{2}x + \bar{2} \in \mathbb{Z}_3[x]$

$B = x^3 + x^2 - x + \bar{1} \in \mathbb{Z}_3[x]$

$g = x^3 - x + \bar{1}$

$g(0) = \bar{1}$, $g(\bar{1}) = \bar{1}$, $g(\bar{2}) = \bar{1}$ $\Rightarrow g$ – неразложим

$\Rightarrow \overline{F}$ – поле с ел.: $\bar{f} = \underset{\wedge\wedge}{a}x^2 + \underset{\wedge\wedge}{b}x + \underset{\wedge\wedge}{c}$ $a, b, c \in \mathbb{Z}_3$

$\overline{A}$: $A: x^4 - \bar{2}x + \bar{2}$

$- x^4 - x^2 + x$

$x^2 - \bar{3}x + \bar{2}$

$\bar{3} = 0$

$g = x^3 - x + \bar{1}$

$g = x$

$q = 1$

$\overline{A} = x^2 + \bar{2}$

$\overline{B}$: $B = x^3 + x^2 - x + \bar{1}$

$- x^3 \quad - x + \bar{1}$

$B = x^2$

$g = x^3 - x + \bar{1}$

$g = \bar{1}$

$q = 2$

$\overline{B}^{-1}$: $(x^3 - x + \bar{1}, \underset{\overline{B}}{x^2}) = 1$

$\forall u, v \in \overline{F}$:

$u \cdot g + v \cdot \overline{B} = \overline{1}$

$\overline{B} v \equiv 1 \pmod{g}$

$g = x^3 - x + \overline{1}$ $\quad\quad$ $\overline{B} = x^2$

$-x^3$ $\quad\quad$ $q_2 = x$

$r_1 = -x + \overline{1}$

$\overline{B} = x^2$ $\quad\quad$ $r_1 = \overline{2}x + \overline{1}$

$-x^2 + \overline{2}x$ $\quad\quad$ $q_3 = \overline{2}x + \overline{2}$

x

$-x + \overline{2}$

$r_2 = \overline{1}$

$\overline{1} = \overline{B} - r_1(\overline{2}x + \overline{2}) = \overline{B} - (g - x\overline{B})(\overline{2}x + \overline{2}) = (x + \overline{1})g + (\overline{1} + \overline{2}x^2 + \overline{2}x)B$

$u = \overline{B}^{-1}$

$\overline{AB^{-1}} = (x^2 + \overline{2}) \cdot (\overline{2}x^2 + \overline{2}x + \overline{1}) = \overline{2}x^4 + \overline{2}x^3 + \overline{2}x^2 + x + \overline{2}$

$\overline{2}x^4 + \overline{2}x^3 + \overline{2}x^2 + x + \overline{2}$ $\quad\quad$ $g = x^3 - x + \overline{1}$

$g = \overline{2}x + \overline{1}$

$\overline{AB^{-1}} = \overline{2}x^2 + 1$ - най е грешен отг.

Зад: $f = x^3 + px + q \in \mathbb{C}$

$g = x^3 + 7x - 6 \in \mathbb{C}$

? $h = x^3 + ax^2 + bx + c \in \mathbb{C}$?

с корени $g(x_1), g(x_2), g(x_3)$, където x_1, x_2, x_3 са корени на f.

Виет за f:

$$\begin{cases} x_1+x_2+x_3=0 \\ x_1x_2+x_1x_3+x_2x_3=p \\ x_1x_2x_3=-q \end{cases}$$

$g(x_i)=y_i,\ i=1,2,3$

Виет за h:

$$\begin{cases} y_1+y_2+y_3=-a \\ y_1y_2+y_1y_3+y_2y_3=b \\ y_1y_2y_3=-c \end{cases}$$

$f(x_i)=0=x_i^3+px_i+q \to x_i^3=-px_i-q,\ i=1,2,3$

$g(x_i)=x_i^3+7x_i+6=(7-p)x_i+6-q,\ i=1,2,3$

$$\sum_{i=1}^{3} y_i=(7-p)\underbrace{(x_1+x_2+x_3)}_{0}+3(6-q)=18-3q=-a \to a=3q-18$$

$$\sum_{1\le i<j\le 3} y_iy_j=(7-p)^2\underbrace{(x_1x_2+x_1x_3+x_2x_3)}_{p}+3(6-q)^2+\cancel{(7-p)(6-q)\cdot 2\cdot\underbrace{(x_1+x_2+x_3)}_{0}}=b$$

$$y_1y_2y_3=(7-p)^3\cdot\underbrace{x_1x_2x_3}_{-q}+(6-q)^3+(7-p)^2\cdot(6-q)\underbrace{(x_1x_2+x_1x_3+x_2x_3)}_{p}+\cancel{(7-p)\cdot(6-q)^2\underbrace{(x_1+x_2+x_3)}_{0}}=-c$$

Зад: $f=x^3+3x+1$

$g=x^3+2x-1$

? $h=x^3+ax^2+bx+c$

с корени $g(x_1), g(x_2), g(x_3)$, —"—

Д.Р.

Зад: $I=(212)\trianglelefteq\mathbb{Z}$ и $J=\{x\in\mathbb{Z}\mid \exists n=n(x)\in\mathbb{N}: x^n\in I\}$

Да се докаже, че J е идеал на $\mathbb{Z}$, съдържащ I, и да се определи ест. т. n: $J=(n)$

$x,y\in\mathbb{Z}$: $\exists u=n(x)$, $m=m(y)$:
$x^n\in I$, $y^m\in I$

1) $x-y\in J \longrightarrow (x-y)^{m+n-1}=\dots\in I$
2) $xr\in J$, $\forall r\in\mathbb{Z}$
$(xr)^n=x^n r^n\in I$

$\Rightarrow J\trianglelefteq\mathbb{Z}$, $I\subset J$

$212=2^2\cdot 53$

$НОД(2,53)=1$

$x\in\mathbb{Z}$: $x\in J \Leftrightarrow^{\exists m;}\ x^m\in I \Leftrightarrow 212\mid x^m \Leftrightarrow 2^2\cdot 53\mid x^m \Leftrightarrow 2\mid x^m$ и $53\mid x^m$
$\Rightarrow 106\mid x^m$
$\Rightarrow x^m\in(106) \Rightarrow J\subseteq(106)$
но $(106)\subseteq J$ $(106^2\in I)$

$\Rightarrow J=(106)$